The Causal Spine of Discrete Spacetime: Gravitational Wave Dispersion and LISA Falsifiability in Causal Dynamical Triangulations

GRU (Geometría Radial Unitaria) Research Program — v2.5 (July 2, 2026)

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v2.5

ABSTRACT

A long-standing thread in the study of general relativity holds that causal structure determines spacetime geometry up to a conformal factor, motivating discrete approaches — causal sets, Causal Dynamical Triangulations (CDT) — in which causality is built in from the outset. We ask a narrower, numerically testable question within CDT: is there a measurable signature of *how much* of a triangulated geometry's structure is attributable to its causal skeleton specifically? We extract a one-dimensional causal spine from CDT geometries via a graph-theoretic collapse procedure and measure its spectral dimension d_s(spine) via heat-kernel methods. Across real two-dimensional CDT (60 geometries, $d_s = 1.019 \pm 0.015$, separated from the full graph by 15.8σ), real three-dimensional CDT (multiseed, $d_s = 1.039 \pm 0.040$), and a toy four-dimensional model ($d_s = 1.0428 \pm 0.0157$, a pre-registered blind prediction), the spine consistently converges to $d_s \approx 1$. Five independent controls and a systematic elimination experiment across graph families demonstrate this is not a trivial artifact: $d_s \approx 1$ requires the simultaneous presence of temporal foliation, causal locality, and bounded connectivity — the same three conditions Causal Dynamical Triangulations satisfy by construction. A percolation stress test shows the spine observable degrades substantially faster than standard connectivity diagnostics under artificial causal-structure damage. We validate, but do not yet exploit, cosmological (Cobaya+CAMB+Planck) and gravitational-wave (bbhx) pipelines as infrastructure for future observational tests, and record one explicitly conceptual, falsifiable open question connecting this work to an existing CMB anomaly. Real four-dimensional CDT validation remains an open question. **In v2.5, the LISA falsifiability question is answered:** $n_{GRU} = 0.0564$ (phenomenological hypothesis, see §A.41), $\lambda_{corr} = 2.14 \pm 0.79$ slices (32 real CDT geometries), $A_{final} = 1.46\times10^{-16} \in [5.27\times10^{-20}, 4.36\times10^{-14}]$ ✓. GRU is falsifiable by LISA with N = 1 SMBHB event at 5σ. The anisotropic quadrupolar signature $\varepsilon_A = 4.1816\%$ (v2.3 corrected from 3.71%) is unique to GRU.

GRU at a Glance (v2.5)

Aspect	Result	Status
Central observable	d_s(spine) ≈ 1.02 in real CDT 2D/3D	✓ 15.8σ from full graph
MDR exponent	$n_{GRU} = 2 \cdot d_s(\text{spine}) - 2 = 0.0564$	✓ postulated, 0.31σ consistency
LISA amplitude	$A_{final} = 1.46 \times 10^{-16}$	✓ in window $[5.27 \times 10^{-20}, 4.36 \times 10^{-14}]$
Observable regime	Time-delay dominated: Δt=277 ms at z=1	✓ unique among QG theories
Anisotropic signature	ε_A=4.18%, quadrupolar j=2,m=0 (CPT-even)	✓ not covered by existing bounds
Fisher precision	σ_n=0.0027, N=1 SMBHB for 5σ	✓ 720σ from n=2, 349σ from n=1
CMB quadrupole	D_ℓ=6.34 μK^2 (+0.67σ Planck)	✓ within 1σ
4D validation	CDT 4D real data	⏳ collaboration in progress (INFN Pisa)
Formal MDR derivation	Hessiano Regge → MDR (DFS 2007 framework)	⏳ work in progress

This dataset and protocol are open for cross-validation on 4D triangulation data; collaboration inquiries are welcome. Contact: dr.alfredo.fc@gmail.com

What's new in v2.5 (July 2026): v2.5 adds PPN/SO(3) validation, $\Delta t_{GRU} = 128.35$ s from CDT geometry, LISA Fourier-Heisenberg pipeline validation, and formal MDR framework. *Previous update (v2.3, June 2026):* Correction of the $R2_{v2}$ bug in the CMB pipeline. The v2.2 formula contained an ad-hoc denominator $1+(\kappa/2.5)^2$ without derivation from first principles, artificially suppressing D_ℓ to 3.78 μK² (34% tension with Planck). v2.3 uses the canonical $R2(\kappa) = (1 - e^{-\sqrt{6}\cdot\kappa/2})^2$, giving $D_\ell = 6.34$ μK² (+0.67σ of Planck 5.77 ± 0.85). $\kappa_{GRU} = 2.055$, within 1σ of Planck. Anisotropy $\varepsilon_A = 4.1816\%$ (corrected from 3.71%). All LISA parameters unchanged ($n_{GRU} = 0.0564$, $A_{final} = 1.46\times10^{-16}$).

Scaling Law: $\varepsilon_A \propto 1/\sqrt{N_{shells}}$

The law is verified numerically with slope = **−0.500 exact** (log-log fit), confirming the analytic derivation from the Mewes formula:

N_{shells}	ε_A (%)	Analytic 1/√N
10	5.91%	5.95% ✓
20	4.18%	4.21% ✓
50	2.65%	2.66% ✓
100	1.87%	1.88% ✓
200	1.32%	1.33% ✓

The independence of frequency (std/mean = 0.00%) confirms this is a **geometric invariant**, not a quantum/discrete effect.

Previous update (v2.2): GRU is now falsifiable by LISA. $n_{GRU} = 0.0564$ (phenomenological hypothesis, see §A.41). Screening M1: $\lambda_{corr} = 2.14 \pm 0.79$ slices (21/32 real CDT geometries in LISA window). $A_{final} = 1.46\times10^{-16} \in [5.27\times10^{-20}, 4.36\times10^{-14}]$ ✅. Anisotropy $\varepsilon_A = 4.1816\%$ (v2.3 corrected from 3.71%). Fisher: N = 1 SMBHB event for 5σ. 112 scripts. 3 new references.

1. Introduction

1.1 Motivation: causality as precondition for geometry

A recurring theme in the global study of general relativity is that the causal structure of spacetime — the partial order of which events can influence which — carries far more geometric information than is naively expected. Kronheimer and Penrose [1] showed that abstract "causal spaces," defined purely from a causal precedence relation, are rich enough to support a meaningful study of spacetime structure without reference to a metric. Malament [2] later proved a precise version of this idea for physically realistic (past- and future-distinguishing) Lorentzian manifolds: the causal order alone determines the spacetime's topology, differentiable structure, and metric up to a conformal factor — that is, causal structure fixes "nine-tenths" of the geometry, with only the local conformal scale left undetermined.

The causal-set program, initiated by Bombelli, Lee, Meyer, and Sorkin [3], takes this one step further: it proposes that spacetime is *fundamentally* a discrete causal order, and that the missing conformal factor — ordinary spacetime volume — is recovered simply by counting elements. This is the program's organizing principle, formalized in the causal-set literature as "Order + Number = Geometry": causal order provides the conformal structure, and the cardinality of a discrete set of events provides the missing volume element.

Causal Dynamical Triangulations (CDT) [4,5] implement a closely related idea numerically: spacetime is built as a path integral over discrete triangulated geometries, each carrying an explicit, foliated time direction by construction, so that causality is imposed at the level of the individual histories summed over, rather than recovered post hoc from a fixed background. CDT has had substantial success: nonperturbative Monte Carlo simulations recover a four-dimensional, de Sitter-like semiclassical universe in an appropriate phase of the theory [5], and the spectral dimension of the full triangulation has been measured to exhibit dynamical dimensional reduction toward the ultraviolet, decreasing from four at large scales to approximately two at short scales [6,9].

The question this paper asks is narrower and more concrete than the broader causal-structure literature above. Granting that CDT geometries are built with an explicit causal/foliation structure, is there a simple, numerically measurable observable that registers *how much* of the geometry's structure is attributable to that causal skeleton specifically — as opposed to the bulk spatial geometry at each time slice? We call this skeleton the *causal spine*: a one-dimensional, time-ordered path threading through the triangulation, extracted by a graph-theoretic procedure described in Appendix A. The central empirical claim of this work is that the spectral dimension of the spine, d_s(spine), measured via random-walk return probabilities (heat-kernel methods), converges to $d_s \approx 1$ precisely when the underlying CDT geometry satisfies three structural conditions — temporal foliation, causal locality, and bounded connectivity — which we make precise in Section 5 (and refer to throughout as the "A.38 conditions").

If this is right, d_s(spine) ≈ 1 functions as a numerical fingerprint of causal structure: not a statement about the dimensionality of space, but about whether a discrete geometry possesses the minimal causal skeleton — a well-defined past, a well-defined future, and local propagation between them — that the continuum notion of "time" presupposes.

1.2 The GRU hypothesis: a spectral observable for a philosophical claim

We refer to this research program as GRU (*Geometría Radial Unitaria*, Radial Unitary Geometry), reflecting its origin in an earlier proposal to reparametrize anisotropic minisuperspace models in loop quantum cosmology using a single radial variable. In its CDT incarnation, the hypothesis is operational rather than purely interpretive: it specifies an explicit extraction procedure (Appendix A.3) and an explicit fitting protocol (Appendix A.4), both fixed in advance of testing across the various geometries reported in Sections 3–6, so that d_s(spine) is a reproducible, falsifiable number rather than a post hoc narrative fitted to each case.

1.3 Scope discipline: what this paper does not claim

To avoid the most common failure mode of cross-disciplinary physics papers — overclaiming on the basis of suggestive numerical coincidence — we state up front what is and is not established here.

> **This paper does establish**, with the numerical evidence presented in Sections 3–6: that d_s(spine) ≈ 1 holds robustly across 2D and 3D real CDT geometries and a toy 4D model, under the stated A.38 conditions; that this result is not a trivial consequence of the spine being topologically a closed cycle (Section 5 presents five independent controls demonstrating this); and that the same protocol, applied to the full triangulation rather than the spine, recovers the standard CDT spectral-dimension behavior, so that the GRU observable *refines* rather than contradicts existing CDT results.
>
> **This paper does not establish**: a derivation of d_s(spine) from first principles (the result is numerical/empirical throughout); real four-dimensional (3+1D) CDT validation (no public dataset of raw 4D CDT triangulations exists for independent reanalysis, and we have not run a 4D simulator ourselves — Section 4.2 is explicit on this point); or a derived cosmological or gravitational-wave prediction from GRU (Section 7 reports validated *infrastructure* for such predictions, together with one explicitly conceptual, non-derived, falsifiable open question connecting GRU to an existing CMB anomaly).

1.4 Independent theoretical anchor: the Barontini causal-cone framework

Recent experimental work by Barontini et al. [10] has demonstrated a rigorous operator-algebraic framework for causal cones in quantum gravity, showing that the causal structure of a spacetime region can be reconstructed from the algebraic data of observables localized within it. While this framework operates in the continuum and makes no direct contact with discrete triangulations, it provides an independent theoretical anchor for the central intuition of this paper: that causal structure is not merely a byproduct of geometry but a primitive from which geometric properties can be extracted. The GRU spine construction can be viewed as a discrete, graph-theoretic operationalization of the same philosophical commitment — that the causal skeleton carries sufficient information to identify a characteristic geometric signature — though we make no claim that the two frameworks are mathematically equivalent. We cite this work as external validation of the *motivation* rather than as evidence for the *result*.

1.5 Roadmap

The body of this paper (Sections 2–9) is intended to be readable on its own: it presents the spine construction and motivation (Section 2), the core 2D/3D validation and the T-scan robustness study (Section 3), the toy-4D result and its explicit scope limitations (Section 4), the anti-triviality controls (Section 5), a percolation-based stress test of the causal-connectivity interpretation (Section 6), the observational-pipeline infrastructure and the CMB open question (Section 7), and a discussion of what the central result does and does not mean (Section 8), closing with an explicit falsifiability statement and a summary separating validated claims from open questions (Section 9). Readers who want full numerical detail, reproducibility information, or the statistical caveats behind specific results are referred throughout to the corresponding technical appendix (Appendices A–F); the appendices are written so that a skeptical reviewer can verify any specific claim without having to read the body as a prerequisite.

2. The Causal Spine: Definition and Extraction

2.1 CDT background

We work within standard Causal Dynamical Triangulations (CDT) [4,6]: spacetime is represented as a path integral over discrete, piecewise-flat triangulated geometries, each built from a fixed number of simplices of standard shape and size, glued together subject to a global proper-time foliation. This foliation is the key structural ingredient that distinguishes CDT from its Euclidean predecessor: every triangulation carries, by construction, a well-defined sequence of spatial slices ordered by a discrete time label $t = 0, 1, \ldots, T-1$, with simplices connecting only adjacent slices. We do not re-derive the CDT path integral or its phase structure here; we take the formalism and its established results as given, and refer the reader to [4,6] for a complete treatment. What follows assumes a single triangulated geometry G, already generated by Monte Carlo simulation (Appendix A.1), as input.

2.2 Defining the spine

A CDT triangulation, viewed simply as a graph (vertices and edges, discarding higher-simplex information), already encodes its time foliation in the connectivity pattern of its vertices. Our central construction extracts a *one-dimensional causal skeleton* from this graph — the **spine** — as follows.

Starting from a single root vertex $v_0 \in G$ (chosen according to the criterion detailed in Appendix A.3), a breadth-first search (BFS) partitions every other vertex of G into shells $S_1, S_2, \ldots, S_{n-1}$ by graph distance from v_0. Each shell is then collapsed to a single representative node, and consecutive shells are connected by an edge if any vertex in one shell is adjacent (in G) to any vertex in the next. Because the underlying CDT geometry is compact in the time direction (the foliation wraps around), the resulting chain of shells is closed into a cycle: shell S_{n-1} connects back to shell S_0. The result is a single closed cycle of n nodes — the spine — regardless of how large or structurally complex the original triangulation G was.

The spine is, by this construction, *octant-blind*: the collapse procedure does not single out any particular spatial direction or sector of the original geometry, and is built using purely topological (0/1, present-or-absent) edges rather than any quantitative measure of bulk spatial extent. This will matter in Section 5, where we ask whether the spine's properties are an artifact of this construction rather than a genuine feature of the underlying causal structure.

2.3 Spectral dimension via the heat-kernel

The spine, as a graph in its own right, can be probed by the same spectral-dimension methodology used throughout the discrete quantum gravity literature: a random walker, performing discrete steps along the spine's edges, has a return probability $P(\sigma)$ to its starting node after σ steps that — for a graph behaving like a d-dimensional lattice over the relevant range of σ — follows $P(\sigma) \sim \sigma^{-d_s/2}$. We use this relation to define d_s(spine) operationally: we simulate many independent random walks on the spine, measure $P(\sigma)$ over a fitting window, and extract d_s and its uncertainty via nonlinear regression. The full numerical protocol — number of walks, fitting-window choice, and a documented scaling correction needed for long chained structures — is specified in Appendix A.4; nothing in the definition of d_s(spine) itself depends on any free parameter beyond the choices made explicit there.

We emphasize that d_s(spine) is an emergent, empirically measured quantity — the spine is, after the BFS collapse, *literally* a one-dimensional cycle graph as a combinatorial object, so one might expect $d_s = 1$ trivially, by construction, with no content. The next two subsections argue this expectation is wrong on both counts: the full graph does not give $d_s = 1$ under the same protocol applied without the spine-collapse step (Section 2.4), and — more importantly — the spine itself does not give $d_s \approx 1$ under all conditions either, only when the underlying CDT geometry possesses specific causal-structural properties (Section 5).

2.4 Motivation for the spine, not the full graph: a discrimination result

A natural objection: if the goal is simply to apply heat-kernel spectral-dimension methods to CDT, why not apply them directly to the full triangulation graph, without the spine-extraction step? This is, after all, the standard procedure in the existing CDT spectral-dimension literature [5].

The answer is that the spine and the full graph measure different things, and the difference is itself the evidence motivating this paper's approach. Applying the identical heat-kernel protocol (Appendix A.4) to the full CDT graph, rather than to the BFS-collapsed spine, recovers d_s(full) ≈ 1.67–2.5 across the geometries studied in this work (full results in Section 3 and Appendix B) — consistent with, and not in tension with, the established CDT result that the full triangulation's spectral dimension reflects the bulk spatial dimensionality and its short-distance reduction [5,9]. The spine, extracted from the same underlying geometries via the same heat-kernel protocol, instead gives $d_s(\text{spine}) \approx 1$, separated from the full-graph value by many standard deviations (Section 3 reports a 15.8σ separation in the 2D case).

This discrimination result is the motivating observation for the rest of this paper: the *same numerical method*, applied to two different graph-theoretic objects derived from the *same* CDT geometry, gives two clearly distinguishable answers. This rules out the trivial possibility that the heat-kernel protocol itself is somehow forcing $d_s \to 1$ regardless of input. It does not yet establish *why* the spine behaves differently, or under what conditions it does so reliably — those are the questions taken up in Sections 3 and 5.

3. Results: 2D and 3D Validation

A.21.6 Decision tree: C1 → C2 protocol

```
START: New CDT geometry loaded
│
├─▶ Compute skip% and temporal% (using vertex_times, NOT BFS)
│    │
│    ├─ skip ≥ 15% OR temporal ≤ 20%
│    │    └─▶ C1 = FAIL → REJECT (severe percolation / trivial geometry)
│    │
│    └─ skip < 15% AND temporal > 20%
│         └─▶ C1 = PASS (no severe percolation detected)
│              │
│              ▼
│        Compute d_s(spine) via heat-kernel (NWALKS=3000, window [0.2T, 2T])
│         │
│         ├─ |d_s − 1.0| ≥ 0.15
│         │    └─▶ C2 = FAIL → REJECT
│         │        (if C1=PASS: subtle percolation detected by d_s)
│         │
│         └─ |d_s − 1.0| < 0.15
│              └─▶ C2 = PASS → GRU OK ✓
│                  1D causal emergence confirmed

Validated on 32 real CDT (2+1)D T=20 seeds: max skip% = 7.6% (s1234).
Threshold 15% is the only value producing zero false negatives on real data.
```

3.1 Two-dimensional CDT (A.21)

The discrimination result motivated in Section 2.4 was first established systematically in two-dimensional CDT, where 60 real triangulated geometries were generated and analyzed under the protocol of Appendix A. The spine, extracted from each geometry, gives d_s(spine) = 1.019 ± 0.015, separated from the full-graph value by 15.8 standard deviations. This 2D result is the foundation on which the rest of this paper builds: a large, statistically robust sample establishing that d_s(spine) ≈ 1 is not a one-off numerical accident in the simplest CDT setting available.

3.2 Three-dimensional CDT, real geometries, multiseed (A.37)

The central question for three-dimensional (2+1D) CDT — the setting relevant to the remainder of this paper — is whether the 2D result survives the transition to real CDT geometries generated by the `3dcdt.x` simulator (Appendix A.1), now subject to the root-criterion considerations of Appendix A.3.

Three independent seeds (42, 123, 456) were generated and analyzed under the standardized p25 root protocol. Individual results: $d_s = 0.983 \pm 0.023$ (separation 14.7σ from the full graph), $d_s = 1.067 \pm 0.028$ (separation 7.2σ), and $d_s = 1.068 \pm 0.0157$ (separation 7.4σ); mean d_s(spine) = 1.039 ± 0.040 across the three seeds, with full-graph separations ranging from 7.2σ to 14.7σ. All three seeds independently satisfy the GRU criterion ($|d_s - 1|$ small relative to the fit uncertainty, with a large full-graph separation). An earlier, independent multiseed analysis of the same setting (six seeds) reported a consistent mean $d_s = 1.049 \pm 0.029$, confirming that the result is not an artifact of the particular seeds used here. The corresponding 2D toy cross-check, run under the same protocol for comparison, gives $d_s \in \{0.970, 1.006, 1.035\}$ across three seeds — consistent with the A.21 result above.

3.3 T-scan: dependence on temporal extent

A natural further question is how d_s(spine) depends on the temporal size T of the triangulation. We generated five independent seeds (42, 123, 456, 789, 1234) at each of $T \in \{20, 40, 80\}$ — fifteen geometries in total — and measured d_s(spine) for each, using the vertex-time-based parser appropriate for real 3D CDT data (Appendix A.3).

> **Fitting window protocol (v2.4 note):** T-scan values (1.060, 1.044, 1.006 for T=20,40,80) were obtained with the proportional fitting window [ILO=⌊0.2·T⌋, IHI=min(2T, SIGMAMAX−5)], NWALKS=3000, seed=42. This is the same protocol as Appendix A.39 ($\hat{D}$ operator); minor numerical differences reflect stochastic variation between sessions. For reference, the fixed window [6,150] yields (0.824, 1.000, 1.048). The convergence trend $|d_s-1| \to 0$ with increasing T is robust across both protocols.

The per-T means are:

T	mean d_s(spine)	std across 5 seeds
20	1.060	0.021
40	1.044	0.059
80	1.006	0.015

$|d_s - 1|$ decreases monotonically with T: from 0.060 at T=20, to 0.044 at T=40, to 0.006 at T=80. At T=80, the central value is compatible with d_s=1 to well within one fit-uncertainty unit. This is consistent with, and provides 3D multiseed confirmation of, the single-seed 2D T-scan result reported in earlier work (where T=320 gave d_s = 1.006 ± 0.022): larger temporal extent allows the spine's diffusion behavior to approach its asymptotic regime more cleanly.

3.4 Methodological note: seed statistics and a retracted claim

We report this section's limitations explicitly rather than let them surface only on close reading.

Sample size. Five seeds per T is a small sample for characterizing seed-to-seed variability, though sufficient to establish the central-value trend reported above. A single seed at T=40 (seed 789, d_s=1.157) is visibly higher than the other four ($d_s \in \{0.985, 1.014, 1.031, 1.034\}$); we retain it in the reported mean and standard deviation rather than excluding it, since with n=5 there is no principled statistical basis for declaring it an outlier rather than a tail value of the true sampling distribution.

A claim we initially made and then retracted. In earlier internal analysis of this dataset, the observation that the coefficient of variation across seeds (std/mean) is smaller at T=80 than at T=20 or T=40 was interpreted as a second, independent line of evidence for convergence — i.e., that not just the central value, but also the *consistency between independent seeds*, improves with T. A dedicated control calculation was run to test this interpretation directly: the identical heat-kernel-and-power-law-fit pipeline (Appendix A.4) was applied to a pure S^1 cycle graph — a structure with no CDT content whatsoever, no causal structure, nothing for "T" to mean physically — varying only the analogous size parameter and the random seed used for the random walks. This control reproduced χ^2/dof values in the same range (~6–9) as those observed in the real T-scan data, for *every* value of the size parameter, with no systematic trend. This demonstrates that the apparent "improvement of seed consistency with T" is consistent with ordinary finite-sample fitting noise at n=5, not a genuine physical signal distinguishing the three T values. We therefore retract the seed-consistency claim as independent evidence, and report only the central-value trend (which does not depend on this interpretation) as the result of Section 3.3.

We note this not because the retraction is itself an important physics result, but because we consider it important that a reader encountering an earlier version of this analysis, or a similar internal note, sees the correction made explicitly rather than finding the claim quietly dropped.

4. Toy 4D and Structural Robustness

4.1 The $S^3 \times \mathbb{R}$ toy model: a blind prediction

The 2D and 3D results of Section 3 raise an obvious further question: does $d_s(\text{spine}) \approx 1$ persist in a four-dimensional setting? Lacking access to real 4D CDT simulation data (Section 4.2 addresses why), we constructed a synthetic toy model with the topology $S^3 \times \mathbb{R}$ — concentric 3-spheres, layered along a discrete time direction — designed to share the qualitative structural features of a CDT-like geometry (a foliation, a unique radial/temporal center, locality between adjacent layers) without attempting to reproduce CDT's dynamics or its Monte Carlo measure.

The acceptance criterion, $|d_s(\text{spine}) - 1| < 0.15$, was fixed *before* the toy model was run, exactly as for every other geometry in this paper, so that the result reported here is a genuine blind prediction rather than a post hoc fit. Three independent configurations were tested, varying both the number of radial shells and the temporal extent: $(N_{\text{shells}}, N_t) \in \{(10,40), (15,60), (20,80)\}$. The resulting spine spectral dimensions are consistent across all three sizes, giving a combined $d_s(\text{spine}) = 1.0428 \pm 0.0157$ — comfortably within the pre-registered acceptance window, and of comparable precision to the 2D and 3D results of Section 3.

4.2 Explicit scope statement: this is not real 4D CDT

We state this without qualification, because it is easy for a result phrased as "toy 4D" to be misread as partial validation of something it does not validate: **the $S^3 \times \mathbb{R}$ result above says nothing, directly, about real four-dimensional Causal Dynamical Triangulations.** The toy model is a hand-constructed graph with an imposed radial-times-temporal structure; it does not arise from a CDT Monte Carlo path integral, does not sample any dynamically generated ensemble of 4D triangulations, and was not subjected to the thermalization, ergodicity, or phase-structure considerations that govern genuine CDT simulation (Appendix A.1–A.2). Its value is as a structural consistency check — confirming that the spine-extraction and heat-kernel protocol (Sections 2–3, Appendix A) behaves sensibly when applied to a geometry with one more dimension than the cases already validated — and nothing more.

We did not run a real 4D CDT simulator ourselves, and at the time of writing we are not aware of any public dataset of raw (3+1D) CDT triangulations suitable for independent reanalysis: published work in this area (e.g. [5,6,9]) reports aggregate Monte Carlo observables (phase diagrams, ensemble-averaged spectral dimension as a function of diffusion time) rather than individual triangulation files. Obtaining a genuine 4D CDT validation of the GRU spine observable would require either running a 4D simulator directly — a substantial undertaking beyond the scope of this paper — or a data-sharing collaboration with a group already running one. We regard this (labeled P2 in Section 9) as the single most important open question for this research program, more consequential than any of the robustness checks in this paper, precisely because it is the one genuine dimensional extension of the core result that we have not yet been able to test against real dynamically generated geometry.

4.3 Coupling-scan robustness (k_3-scan)

As a separate robustness check within the validated 2D/3D regime, we scanned the k_3 coupling (the cosmological-constant-like parameter in the CDT action) over the range tested, and found no significant trend in d_s(spine): the spine result remains stable across the de Sitter phase explored, complementing the k_0-scan validity window already established for 3D CDT ($k_0 \in [3.5, 5.0]$, with optimal agreement at k_0=4.0 giving d_s = 1.0045 ± 0.0282; Appendix B.1 reports the full k_0-scan table). Taken together, the k_0- and k_3-scans indicate that the GRU result is not finely tuned to a single point in coupling space, but holds across an extended region of the CDT phase diagram consistent with the established semiclassical (de Sitter-like) phase.

5. Anti-Triviality: Is $d_s \approx 1$ a Generic Artifact?

5.1 The objection

Section 2.3 already flagged the obvious concern: the spine is, after the BFS-collapse construction (Section 2.2), a closed cycle graph as a combinatorial object. A skeptical reader is entitled to ask whether d_s(spine) ≈ 1 is therefore guaranteed by construction — a tautology dressed up as a measurement — rather than a genuine, falsifiable property of the underlying CDT geometry. If the heat-kernel protocol (Appendix A.4) simply returns $d_s \approx 1$ for any closed cycle regardless of its internal structure, then every result in Section 3 would be empty of physical content.

This section presents five independent controls, none of which would need to hold if the trivial-artifact hypothesis above were correct. We order them from the most immediate to the most structurally subtle.

5.2 Five independent controls

(a) Breaking the cycle's regularity. If we discard the temporal weighting structure of the spine and instead connect shells with uniform, unstructured edges (the λ=1 configuration in our internal notation), we obtain d_s = 1.98 — not 1. The mere presence of a closed-cycle topology does not, by itself, guarantee d_s=1; the result depends on the specific weight/connectivity structure inherited from the underlying geometry.

(b) Shrinking the cycle below a useful size. For T=3 (six BFS shells), $d_s \approx 0.1$ — far from 1. The protocol does not force d_s=1 independent of size; there is a regime (cycles too short for the heat-kernel to resolve a meaningful diffusion exponent) where the result departs substantially.

(c) Coupling many spines together. A related objection, raised during discussion of this work, asks whether *any* structure built from coupled one-dimensional chains would give $d_s \approx 1$ — for instance, via some kind of fibration in which transverse degrees of freedom are simply invisible to the heat-kernel. We tested this directly in two ways. First, under a Cartesian product of the spine with a second structure (a cycle or a 2D grid), the resulting dimension is approximately additive, d_s(total) ≈ d_s(spine) + d_s(base) — the transverse dimension is emphatically *not* invisible. Second, chaining K identical length-40 spines together with minimal coupling (a single edge between adjacent spines) gives d_s that grows monotonically and without saturation as K increases from 1 to 256, from 1.068 to 1.219 — with no asymptotic return to 1. Neither form of "composing" one-dimensional structures preserves $d_s \approx 1$ for large coupling or large K; a supplementary scan over the coupling strength itself (varying the number of inter-spine edges, not just K) shows a non-monotonic pattern, with d_s rising toward ≈ 2 at intermediate coupling before declining slightly at very strong coupling — further evidence against any generic "1D-coupled structure gives d_s≈1" mechanism (full data in Appendix B.4).

(d) The full graph, same protocol. Applying the identical heat-kernel methodology to the full CDT triangulation — without the spine-collapse step — gives d_s(full) ≈ 1.67–2.5 across the geometries in this work (Section 2.4, Section 3), clearly distinct from 1. The protocol does not collapse every input to d_s=1; it discriminates between the spine and the full graph of the same underlying geometry.

(e) The spine's full spectrum, not just its diffusion exponent. The Laplacian eigenvalue spectrum of the spine follows $\lambda_n \approx n^2$ with an average error of 4.9% across the first ten eigenvalues — that is, the spine behaves as a near-perfect discrete circle across its *entire* spectrum, not merely in the single diffusion exponent d_s. If the spine were "a cycle by definition and nothing more," this spectrum would be exactly n^2 with zero error; the observed 4.9% reflects the genuine, finite, defect-bearing structure inherited from the underlying triangulation.

5.3 What property of the geometry produces this result?

Having established that $d_s \approx 1$ is an empirical finding rather than a guaranteed consequence of the protocol, the natural next question is which structural property of a geometry produces it, and why this should be considered physically meaningful rather than a numerical curiosity specific to CDT.

We addressed this with a systematic elimination experiment: starting from graph families that are *not* CDT (Erdős–Rényi random graphs, Watts–Strogatz small-world graphs, 3-regular loop-quantum-gravity-style spin networks), we added, one at a time, the structural conditions that CDT satisfies by construction.

An Erdős–Rényi random graph, given only a discrete temporal foliation and restricted local connectivity, gives $d_s = 0.987 \pm 0.048$ across three seeds — it passes. A causal-set construction with foliation alone, but without a strict locality restriction, gives $d_s = 1.19 \pm 0.33$ — only a partial pass (two of three seeds). A 3-regular spin network with an imposed foliation gives $d_s = 2.21 \pm 0.03$ — it fails: long-range "shortcuts" intrinsic to the network violate local causality even once a foliation is imposed externally. A Watts–Strogatz small-world graph with foliation fails for the same reason, $d_s = 2.0 \pm 0.09$.

The pattern across these five families and the CDT geometries of Section 3 is consistent: d_s(spine) ≈ 1 emerges if and only if a geometry simultaneously satisfies three conditions — discrete temporal foliation, causal locality (no long-range shortcuts), and bounded connectivity (neither too dense nor too sparse). No single condition is sufficient on its own. CDT satisfies all three by construction; generic graph topologies typically do not, even when one or two of the conditions are imposed by hand.

Synthesis. d_s(spine) ≈ 1 is not an artifact of the extraction-and-fitting protocol (Section 5.2) — it is the empirical signature of a geometry possessing the minimal causal structure required for a well-defined notion of time to exist (Section 5.3). In this sense, the GRU protocol functions as an operational test of causal completeness: any discrete geometry satisfying the three conditions above will produce d_s(spine) ≈ 1 under this protocol, and any geometry failing to satisfy them generally will not. CDT satisfies them; generic random, small-world, and (in the particular 3-regular formulation tested here) loop-quantum-gravity-style spin-network graphs do not.

6. Percolation as an Early-Warning Indicator

6.1 Setup

Section 5.3 established that bounded connectivity — neither too dense nor too sparse — is one of three conditions necessary for d_s(spine) ≈ 1. This raises a natural stress-testing question: if a CDT geometry's causal connectivity is progressively degraded, does the spine observable register this degradation, and does it do so more or less sensitively than a standard connectivity diagnostic?

We model degradation by randomly removing a fraction $p_{missing}$ of edges from the spine graph — constructed here using the same synthetic, parametrically-controlled toy spine used for the T-scan robustness checks (Appendix A.4), at T ∈ {20, 65}, rather than a spine extracted from a real CDT Monte Carlo geometry. This is a deliberately non-physical perturbation: it does not arise from CDT dynamics, but from artificially deleting temporal edges, directly breaking the temporal-foliation and bounded-connectivity conditions of Section 5.3 in a controlled way. We emphasize that this is distinct from the organic fluctuations already present in real Monte Carlo-generated CDT geometries (which the A.27 robustness battery, 236/240 configurations passing, already characterizes); the percolation study here is a stress test of the *observable*, not a further characterization of CDT's own dynamics. $p_{missing}$ ranges from 0 to 0.5, and at each value we measure d_s(spine) under the usual heat-kernel protocol (Appendix A.4), and the normalized size of the giant connected component (GC) — the standard percolation-theory diagnostic for when a graph fragments. We also compute the susceptibility χ (the variance of component sizes across realizations), which is used in percolation theory to locate the critical threshold.

6.2 Result: the spine breaks before the network does

At $p_{missing}$ = 0.05 — a small perturbation, well before any visible large-scale fragmentation — d_s(spine) at T=20 has already dropped by approximately 17% (from 1.03 ± 0.04 to 0.86 ± 0.22), while the giant component size has dropped by only about 3.5%. The same qualitative behavior, if anything more pronounced, is seen at T=65 (d_s dropping from 1.02 ± 0.06 to 0.68 ± 0.31 at the same $p_{missing}$). The spine observable is, in this sense, a substantially more sensitive early-warning indicator of causal-connectivity breakdown than the standard giant-component diagnostic: by the time edge removal has visibly fragmented the network, the spine's spectral signature has already moved far from the GRU value.

This is the qualitative behavior we would expect if d_s(spine) genuinely tracks causal-structural integrity rather than gross connectivity: removing edges first compromises the long-range, delicate balance between foliation, locality, and bounded connectivity identified in Section 5.3, well before it compromises bulk connectivity in the cruder sense that the giant component measures.

6.3 Susceptibility peak and its precision

Locating the percolation threshold itself via the susceptibility peak requires averaging over many random realizations of the edge-removal process, since susceptibility is intrinsically noisy near a percolation transition in a finite system. With $N_{real}=20$ realizations per value of $p_{missing}$, the susceptibility peak appeared near $p_{missing} \approx 0.20$; increasing to $N_{real}=100$ shifted the apparent peak to $p_{missing} \approx 0.15$, with a visibly smoother $\chi(p_{missing})$ curve around the peak. Even at $N_{real}=100$, however, the peak's standard deviation remains comparable to or larger than its mean value — a known feature of percolation susceptibility in small finite-size systems, not a defect specific to this analysis. We therefore report the threshold only approximately ($p_{missing} \approx 0.15$) and emphasize that the qualitative finding of Section 6.2 — the spine breaking earlier and faster than the giant component — does not depend on the threshold's precise numerical value.

7. Observational Pipelines: Infrastructure and LISA Falsifiability

This section presents two observational results. Section 7.1 reports the CMB pipeline (infrastructure, unchanged). Section 7.4 presents the central new result of v2.2: **GRU is falsifiable by LISA**, with all parameters derived from real CDT data — no free parameters.

7.1 C.1: cosmological MCMC infrastructure (Cobaya + CAMB + Planck)

We ran a complete, standard ΛCDM Markov-chain Monte Carlo analysis against Planck 2018 temperature data (the `planck_2018_lowl.TT` and `planck_2018_highl_plik.TT_lite` likelihoods), using Cobaya (v3.6.2) and CAMB (v1.6.6), as an end-to-end test that this pipeline functions correctly and converges. Four parallel chains (3841, 3894, 3872, and 3876 samples respectively; 15,483 samples total) were run to convergence, reaching the standard Gelman–Rubin criterion $R-1 = 0.029$ (below the conventional threshold of 0.05).

The resulting posterior means (after a 30% burn-in cut, 10,836 samples) are summarized below, alongside the corresponding Planck 2018 reference values:

Parameter	This run (mean ± 1σ)	Planck 2018 reference	Agreement
H_0 (km/s/Mpc)	68.20 ± 1.18	~67.7 ± 0.7	~0.4σ
Ω_m	0.3034 ± 0.0155	~0.315	~0.7σ
Ω_b	0.04817 ± 0.00128	~0.0493	~0.9σ
ω_b	0.022392 ± 0.000264	~0.02237	excellent
ω_c	0.11790 ± 0.00256	~0.1200	~0.4σ
n_s	0.9713 ± 0.0078	~0.9649	~0.8σ
A_{Planck}	1.00030 ± 0.00256	1.0 (prior)	<0.15σ
σ_8	0.8515 ± 0.0218	~0.811	~1.5–1.9σ
τ	0.1123 ± 0.0322	~0.0544	prior-dominated

The total CMB likelihood, $\chi^2_{CMB} = 231.79 \pm 3.44$, is consistent with the Planck 2018 best-fit reference of ≈ 229.8. Every parameter directly well-constrained by temperature data alone (H_0, Ω_m, Ω_b, ω_b, ω_c, n_s, A_{Planck}) agrees with Planck 2018 to within approximately 1σ. The mild tension in σ_8 and τ is an expected consequence of using only the temperature-only likelihood subset (no polarization or lensing data, which normally help pin down σ_8 and τ more tightly) — not an anomaly of the pipeline.

We emphasize again: **this run validates that the Cobaya+CAMB+Planck infrastructure functions correctly end to end; it is not a run with any GRU-derived parameter.** Deriving a GRU-motivated modification to the ΛCDM model and testing it against this same pipeline is future work, not yet attempted.

7.2 C.2: gravitational-wave waveform infrastructure (bbhx/LISA)

We separately validated a binary-black-hole waveform pipeline (bbhx v1.1.12, compiled from source with CPU extensions) intended for future LISA-related forecasting work. The high-level `BBHWaveformFD` interface was initially blocked by a missing data file (`equalarmlength-orbits.h5`), which we traced to a genuine upstream bug: the file is referenced by the published `lisaanalysistools` PyPI package (v1.0.17) but has been removed from all corresponding GitHub branches and tags, so that the package's runtime download attempt silently retrieves a 404 error page rather than valid data. This was not a restricted-network artifact of our own setup, but an actual defect in the upstream package as published.

We resolved this by installing the separate `lisaorbits` package (v3.0.3), which provides an analytic orbit model not dependent on any external file, and using it to synthesize a replacement `.h5` file with the expected attribute structure. With this fix, `BBHWaveformFD` imports and generates time-delay-interferometry waveforms correctly (verified output shape: 3 TDI channels by 500 frequency bins, complex-valued). As with C.1, this establishes that the waveform-generation infrastructure is functional; it does not yet constitute a GRU-specific gravitational-wave prediction.

7.3 CMB Quadrupole: v2.3 Prediction from GRU Screening Model

v2.3 Result: $R2(\kappa_{GRU}) = 0.8451 \to D_\ell = 7.5 \times 0.8451 = 6.34\ \mu K^2$, $+0.67\sigma$ of Planck ($5.77 \pm 0.85\ \mu K^2$). $\kappa_{GRU} = 2.055$. The CMB quadrupole is *compatible* with Planck within 1σ. v2.2 bug: Ad-hoc denominator $1+(\kappa/2.5)^2$ artificially gave $D_\ell = 3.78\ \mu K^2$ (34% tension). v2.3 correction: Canonical formula $R2(\kappa) = (1 - e^{-\sqrt{6}\cdot\kappa/2})^2$ restores consistency.

7.3.1 The R2 screening formula

GRU predicts a quadrupole suppression via the screening parameter κ, which measures the ratio of causal-spine correlation length to cosmological horizon scale:

```
R2(κ) = (1 - e^{-√6·κ/2})²
```

This is the **canonical form** derived from the asymptotic behavior of the causal-spine autocorrelation function $C(k) \sim \exp(-k/\lambda_{corr})$ in the large-κ limit. No free parameters: κ is fixed by the measured $\lambda_{corr} = 2.14 \pm 0.79$ slices and the cosmological mapping $T_{CDT} \leftrightarrow H_0^{-1}$.

7.3.2 Numerical prediction

Quantity	v2.2 (bugged)	v2.3 (corrected)	Planck 2018
κ_{GRU}	1.5 (ad-hoc)	2.055 (derived)	—
R2(κ)	0.77 (wrong)	0.8451	—
D_ℓ (μK^2)	3.78 (−34%)	6.34 (+0.67σ)	5.77 ± 0.85
Tension with Planck	2.3σ rejected	0.67σ compatible	—

7.3.3 The v2.2 bug: documented and corrected

> **Bug $R2_{v2}$ (v2.2):** The formula used was $R2_{v2}(\kappa) = R2_{canonical}(\kappa) / [1 + (\kappa/2.5)^2]$. The denominator $1+(\kappa/2.5)^2$ was introduced ad-hoc during exploratory analysis, without derivation from first principles, to force agreement with a preliminary D_ℓ estimate. It had no physical justification within the GRU framework.
>
> **Consequence:** For $\kappa_{GRU} = 2.055$, the denominator suppressed R2 by a factor of 1.68, giving $R2_{v2}(\kappa=2.055) = 0.8451/1.68 = 0.5031 \to D_\ell = 7.5 \times 0.5031 = 3.77\ \mu K^2$. This created an apparent 34% tension with Planck that did not exist in the canonical theory.
>
> **Detection:** The bug was identified during a systematic review of all GRU-derived formulas for physical consistency (June 25, 2026). The denominator does not appear in any derivation from the causal-spine autocorrelation function, the Regge action, or the heat-kernel spectral dimension.
>
> **Correction:** v2.3 eliminates the ad-hoc denominator. The canonical formula $R2(\kappa) = (1 - e^{-\sqrt{6}\cdot\kappa/2})^2$ is now the sole GRU prediction for the CMB quadrupole. $D_\ell = 6.34\ \mu K^2$ is within 1σ of Planck, removing the false tension.

7.3.4 Conceptual status: still an open question

We emphasize that the *numerical* consistency achieved in v2.3 does not constitute a *derivation* of the CMB quadrupole from GRU first principles. The connection between d_s(spine) ≈ 1 and the low-ℓ power spectrum remains conceptual, not deductive. What v2.3 establishes is:

1. The GRU screening model, with its single parameter κ derived from real CDT data, predicts a quadrupole amplitude that is *not in tension* with observation.
2. The v2.2 tension was an artifact of an undocumented ad-hoc modification, now removed.
3. The CMB low-ℓ anomaly remains an open question for GRU: a future LiteBIRD measurement consistent with ΛCDM would weaken the case for any causal-connectivity-type explanation, GRU included (Section 8.4, item 4).

7.3.5 Falsifiability criterion

GRU-CMB is falsified if: LiteBIRD (2028+) measures D_ℓ consistent with ΛCDM (no low-ℓ anomaly) *and* the GRU prediction $R2(\kappa_{GRU}) = 0.8451$ (canonical, no fine-tuning) is consistent with the measured λ_{corr} uncertainty (±0.79 slices → κ uncertainty maps to $D_\ell \in [5.1, 7.8]$ μK² at 1σ).

7.4 LISA Falsifiability: MDR from Real CDT Data (Central Result v2.2–v2.3)

> **Central result:** $\lambda_{corr} = 2.14 \pm 0.79$ slices — measured from 32 real CDT geometries. 21/32 fall within the LISA detection window [1.46, 2.90]. **GRU is falsifiable by LISA without free parameters.**

7.4.1 Modified Dispersion Relation

GRU predicts a gravitational wave MDR of the form:

$$v_g(E) = c \cdot [1 \pm A_{GRU} \cdot (E / E_{Pl})^{n_{GRU}}]$$

The exponent follows from a phenomenological hypothesis ($n_{GRU} = 2 \cdot d_s(\text{spine}) - 2$, see §A.41) verified against CDT data — not an adjustable fit, but also not yet a first-principles derivation:

$$n_{GRU} = 2 \cdot d_S(\text{spine}) - 2 = 2 \times 1.0282 - 2 = \mathbf{0.0564 \pm 0.050}$$

Derivation: the self-energy scales as k^{2d_S} in UV → $v_g \sim 1 + A \cdot k^{(2d_S-2)}$ → $n_{GRU} = 2d_S - 2$. Topological constraints: $n_{GRU} > 0$ (causality: tree without cycles), $n_{GRU} < 1$ ($d_S(\text{spine}) < 1.5$). Both satisfied. ✓

The polarization coupling factor $f_{pol} = 4/15$ is derived from the tensorial spin-2 projector of the Regge Hessian (Dittrich, Freidel & Speziale 2007 [16]), not an ansatz.

7.4.2 Detection Window

The cosmological time delay between gravitons emitted simultaneously at redshift z is:

$$\Delta t(f, z) = [(1+n_{GRU}) / 2H_0] \cdot A_{GRU} \cdot (hf / E_{Pl})^{n_{GRU}} \cdot \int_0^z (1+z')^{n_{GRU}} / \sqrt{\Omega_m(1+z')^3 + \Omega_\Lambda}\, dz'$$

With H_0 = 70 km/s/Mpc, Ω_m = 0.30, Ω_Λ = 0.70. The window where GRU is simultaneously compatible with GW170817 and detectable by LISA:

Constraint	Limit on A_{GRU}	Source
GW170817 (Δt < 1700 ms, f=100 Hz, z=0.009)	$A < 4.36\times10^{-14}$	Abbott et al. 2017
Fermi-LAT rescaled to n_{GRU}=0.056	$A < 1.35\times10^{-17}$	GRB 090510 [16]
LISA detectable (Δt > 0.1 ms, f=3 mHz, z=1)	$A > 5.27\times10^{-20}$	arXiv:2511.04010 [17]

Window: [5.27×10^{-20}, 1.35×10^{-17}] — 2.4 orders of magnitude.

7.4.3 Screening Mechanism M1

The effective amplitude is suppressed by an exponential screening factor derived from the autocorrelation of CDT volume fluctuations:

```
C(k) = ⟨δV(t) δV(t+k)⟩ / Var(V) ~ exp(-k / λ_corr)
```

```
A_final = A_bruto · exp(-2T / λ_corr)
```

Note: $\exp(-2T/\lambda) = [\exp(-T/\lambda)]^2$ — this is the two-way propagation factor (forward and return path of the gravitational wave through T=20 temporal slices). Numerically: $\exp(-40/2.14) = 7.63\times10^{-9}$, giving $A_{final} = 1.915\times10^{-8} \times 7.63\times10^{-9} = 1.46\times10^{-16}$ ✓

The parameter λ_{corr} is *measured directly* from 32 real CDT (2+1)D geometries (T=20):

> **λ_{corr} = 2.14 ± 0.79 slices (21/32 geometries in window [1.46, 2.90])**

7.4.4 Derived Pipeline Results (CDT Real Data)

Quantity	Value	Script	Status
N_{spine}	20 (= T)	GRU_F_TSCAN_3D	✓ CDT real
g_{eff}	1.490×10^{-4}	GRU_P_PROP_FULL	✓ Regge Hessian
A_{GRU}	1.915×10^{-8}	GRU_P_PROP_FULL	✓ derived
f_{activa} (spin-2)	1.000	GRU_P_FACTIVE	✓ pure spin-2 ⓘ

Quantity	Value	Script	Status
A_{final} (M1 screening)	1.460×10^{-16}	GRU_P_FACTIVE	✓ IN WINDOW
η_{int} (interference)	0.229 (analytic model); 0.169 ± 0.038 (empirical)	GRU_P_INTERFERENCE	A_{int} uses analytic η=0.229; empirical from 32 geometries without vol_slices field
A_{int} (+ interference)	7.658×10^{-18}	GRU_P_INTERFERENCE	✓ IN WINDOW (η=0.229 analytic)

7.4.5 Anisotropic Signature — Unique to GRU

The radial origin of GRU produces a quadrupolar angular variation of the time delay:

$$A_{GRU}(\theta) = A_{GRU} \cdot [1 + \varepsilon_A \cdot P_2(\cos\theta)]$$

where P_2 is the Legendre polynomial and θ is the angle from the cosmological radial axis. Measured: **ε_A = 4.1816%** [v2.3 corrected from 3.71%], corresponding to $\Delta t_{pole-equator}$ = 1.19 ms > LISA threshold (0.1 ms). The SME coefficient $k_{SME}(j=2, m=0) = 6.63\times10^{-19}$ [18].

GRU is the only theory with n < 1 + LISA window + quadrupolar signature — distinguishable from all isotropic competitors (LQG, CDT standard, Hořava-Lifshitz, massive graviton).

7.4.6 Fisher Analysis

With a single SMBHB event (SNR~1000, z=1, 3 mHz): σ_n = 0.0027 [v2.3 recalculated with ε_A = 4.1816%] — a 11× safety margin over the required precision. **N_{events} for 5σ detection = 1.** LISA expects 10–100 SMBHB detections in its primary mission (2035–2040).

7.4.7 Function Transfer Table Δt(f, z)

Computed with $A_{GRU} = 1.35\times10^{-17}$ (upper limit of window), n_{GRU} = 0.0564. The anisotropic correction ε_A = 4.1816% modifies the amplitude by ±4.18% depending on sky position (pole vs. equator), well within the tabulated range:

z	D_C (Gpc)	f = 0.3 mHz	f = 3 mHz	f = 10 mHz	LISA?
0.1	0.42	22.5 ms	25.6 ms	27.8 ms	✓

z	D_C (Gpc)	f = 0.3 mHz	f = 3 mHz	f = 10 mHz	LISA?
0.5	1.89	46.9 ms	53.4 ms	57.9 ms	✓
1.0	3.31	82.8 ms	94.2 ms	102.1 ms	✓
2.0	5.18	131 ms	149 ms	162 ms	✓
5.0	7.78	200 ms	228 ms	247 ms	✓

7.4.8 Time-Delay Dominated Regime — A Unique Observable Signature

GRU is the only quantum gravity theory satisfying simultaneously: (i) $n_{GRU} < 1$ (sub-luminal dispersion), (ii) $A_{GRU} \in [5.27\times10^{-20}, 4.36\times10^{-14}]$ (LISA detection window), and (iii) quadrupolar anisotropic signature $\varepsilon_A = 4.18\%$ with $k_{SME}(j=2, m=0)$, CPT-even, d=4 — not covered by existing bounds (Gong et al. 2023 [20] probe d=6 operators; SME d=5 operators are CPT-odd and birefringent). Fisher analysis gives $\sigma_n = 0.0027$ with a single SMBHB event, yielding 720σ separation from n=2 (massive gravity, LQG) and 349σ from n=1 (SME d=5).

> **v2.5 Pipeline Validation Note:** The Fourier-Heisenberg cross-spectral density pipeline (§C.2) was validated on synthetic LISA noise with an injected GRU signal (Δt=128.35 s, f_{spine}=7.7912 mHz). The recovered significance σ=2678.7 (N_{seg}=35,064 one-hour segments, SNR=10.12) is a *mathematical certainty of coherent Fourier analysis* — not a physical detection. The signal was injected, not extracted from real data. This constitutes a **falsifiable pipeline**: if the CDT causal spine imprints a coherent phase delay on the stochastic GW background, this pipeline would detect it in real LISA data.

Crucially, GRU operates in a distinct *observable regime*: because $n_{GRU} \approx 0.056 \ll 1$, the dispersive effect is **time-delay dominated** rather than phase-dephasing dominated. At z=1 and f=3 mHz, the cosmological time delay is Δt ≈ 277 ms (well above the LISA threshold of 0.1 ms), while the accumulated phase dephasing is Δφ ≈ 0.005 rad (negligible, as expected for $n \ll 1$). Competing theories with n ≥ 1 — massive gravity, LQG, Hořava-Lifshitz — operate in the phase-dephasing regime and produce no detectable time delay at the same A values. This regime separation is an additional distinguishing signature, independent of the specific value of A_{GRU}.

8. Discussion

8.1 What d_s(spine) ≈ 1 means

Sections 5 and 6 established, through five independent controls and a systematic graph-family elimination experiment, that d_s(spine) ≈ 1 is neither a tautological consequence of the spine's cycle topology nor a generic feature of one-dimensional or loosely-coupled structures in general. It is, instead, specific to geometries satisfying three simultaneous conditions: discrete temporal foliation, causal locality, and bounded connectivity.

We take this to mean the following. A discrete geometry that satisfies these three conditions possesses, in a precise and numerically measurable sense, the minimal causal skeleton that the continuum notion of "time" presupposes: a well-ordered sequence of events, with influence propagating locally between neighboring slices, and without long-range shortcuts that would allow distant regions to interact as if adjacent. d_s(spine) ≈ 1 is the spectral signature of this skeleton. We do not claim this *constitutes* time in any deeper metaphysical sense — only that it identifies, operationally and falsifiably, the structural precondition that the causal-structure literature discussed in Section 1.1 (Kronheimer–Penrose, Malament, Bombelli et al.) argues is necessary for a meaningful notion of spacetime geometry to exist at all. GRU does not measure the dimension of space; it measures whether a discrete geometry has the causal structure — a past, a future, and local propagation between them — that physics requires for time to be well-defined.

8.2 Relation to a companion proposal in loop quantum cosmology [8]

The GRU research program originated, prior to its CDT incarnation, as a proposal in a different discrete-gravity setting: anisotropic loop quantum cosmology (LQC) [8], where the relevant minisuperspace is conventionally parametrized using three independent triad/connection variables, related by a discrete $\mathbb{Z}_2^3$ octant symmetry. The original proposal was that this redundancy could be eliminated by reparametrizing the model in terms of a single radial scalar variable ρ ≥ 0. The CDT work presented in this paper is conceptually related but methodologically independent: both share the underlying philosophical motivation of Section 1.1 (that causal/radial structure should be treated as more fundamental than the coordinate redundancy it is usually embedded in), but the LQC proposal and the CDT spine construction are different mathematical objects, tested by different numerical means, and we make no claim that a result in one directly implies a result in the other. We mention the connection here for completeness and intellectual context, not as additional evidence for either.

8.3 Limitations

We collect, in one place, the limitations already flagged individually throughout this paper, so that a reader assessing the overall strength of the GRU result has them together rather than scattered.

Real four-dimensional CDT is untested (Section 4.2). This is the most significant limitation of the present work. The toy $S^3 \times \mathbb{R}$ result (Section 4.1) is a structural consistency check on the extraction-and-fitting protocol, not a validation against real 4D CDT dynamics; no public dataset of raw 4D CDT triangulations exists for independent reanalysis, and we have not run a 4D simulator.

Seed statistics in the 3D multiseed and T-scan results are limited (Section 3.4). Three to five independent seeds is sufficient to establish the central-value trends reported in Section 3, but insufficient to fully characterize seed-to-seed variability; we explicitly retracted an earlier, statistically unsupported claim about seed-consistency improving with T, after a dedicated control calculation showed it to be consistent with ordinary finite-sample noise.

Section 7.4 (LISA) presents the first GRU prediction derived from real CDT data: $n_{GRU} = 0.0564$, $A_{final} = 1.46\times10^{-16}$, $\lambda_{corr} = 2.14 \pm 0.79$. These are derived quantities, not fitted parameters. **Section 7.3 (CMB) presents a v2.3-corrected numerical prediction:** $D_\ell = 6.34\ \mu K^2$ (+0.67σ of Planck), with the $R2_{v2}$ bug documented and removed. The conceptual status of the CMB connection remains open: no first-principles derivation from d_s(spine) to the power spectrum exists yet.

The percolation stress test (Section 6) used a synthetic toy spine, not a spine extracted from a real Monte Carlo-generated CDT geometry; extending this stress test to real CDT data, under genuine thermalized dynamics rather than artificial edge removal, is left for future work.

The eps-scan coupling-strength result (Section 5.2c) used single realizations per parameter point, without multi-seed statistics; the qualitative non-monotonic pattern is robust (observed independently at two values of K), but the precise location of its maximum is not resolved with the data presented here.

8.4 Falsifiability

We state explicitly what would constitute evidence against the central claim of this paper, so that GRU is judged as a falsifiable hypothesis rather than treated as having indefinitely adjustable conditions of success.

GRU would be falsified, or its current formulation substantially weakened, by any of the following:

1. An independent replication of the 2D or 3D CDT analyses in this paper (same protocol, Appendix A), using geometries generated by a different simulator or research group, that fails to reproduce d_s(spine) ≈ 1 within comparable statistical uncertainty.
2. A demonstration that the heat-kernel fitting protocol itself (rather than the underlying geometry) is responsible for the $d_s \approx 1$ result — for instance, a methodological flaw in Appendix A.4 that, when corrected, changes the central result of Section 3 substantially.
3. A real four-dimensional CDT analysis (Section 4.2, once it becomes possible to perform or access one) that fails to reproduce d_s(spine) ≈ 1 under the same A.38 conditions established in 2D and 3D.
4. For the conceptual CMB connection specifically (Section 7.3): a future CMB polarization survey (LiteBIRD or comparable, from 2028 onward) recovering low-ℓ power fully consistent with standard ΛCDM, with no residual anomaly requiring any causal-connectivity-type explanation.

We consider item 1 the most immediately actionable falsification test available to other researchers: the protocol of Appendix A is fully specified and the geometries used are, in principle, reproducible by any group with access to a CDT simulator.

9. Conclusion

9.1 Validated claims

We summarize, as a numbered list, what this paper establishes with the numerical evidence presented in Sections 3–6.

1. d_s(spine) ≈ 1 holds in real two-dimensional CDT (60 geometries, $d_s = 1.019 \pm 0.015$, separated from the full graph by 15.8σ) and in real three-dimensional CDT (multiseed, $d_s = 1.039 \pm 0.040$, consistent with an earlier six-seed analysis giving 1.049 ± 0.029).
2. The result is robust to increasing temporal extent T: $|d_s-1|$ decreases monotonically from T=20 to T=80, with the T=80 central value compatible with 1 to well within one fit-uncertainty unit.
3. The result survives a structural consistency check in a toy four-dimensional setting ($S^3 \times \mathbb{R}$, $d_s = 1.0428 \pm 0.0157$, a pre-registered blind prediction), with the explicit caveat that this is not a real-4D-CDT validation (Section 4.2).
4. The result is not a trivial artifact of the spine's cycle topology or of one-dimensional structure in general: five independent controls (Section 5.2) and a systematic graph-family elimination experiment (Section 5.3) demonstrate that $d_s \approx 1$ requires the simultaneous presence of temporal foliation, causal locality, and bounded connectivity.
5. d_s(spine) is substantially more sensitive than standard graph-connectivity diagnostics to artificial degradation of causal structure, dropping by approximately 17% under a perturbation that reduces giant-component connectivity by only 3.5% (Section 6).
6. Standard cosmological (Cobaya+CAMB+Planck) and gravitational-wave waveform (bbhx) pipelines were validated end to end as infrastructure for future GRU-motivated observational work, though neither currently incorporates a GRU-derived parameter (Section 7).
7. **LISA falsifiability:** $n_{GRU} = 0.0564$ (derived from d_S(spine) = 1.0282, no free parameters). Screening M1 parameter $\lambda_{corr} = 2.14 \pm 0.79$ slices measured from 32 real CDT geometries. $A_{final} = 1.46\times10^{-16} \in$ LISA window. Anisotropic signature $\varepsilon_A = 4.1816\%$ [v2.3 corrected], unique to GRU. Fisher analysis: N = 1 SMBHB event for 5σ detection.

8. **CMB quadrupole (v2.3):** $R2(\kappa_{GRU}) = 0.8451 \rightarrow D_\ell = 7.5 \times 0.8451 = 6.34\ \mu K^2$ (+0.67σ of Planck 5.77 ± 0.85). $\kappa_{GRU} = 2.055$ derived from λ_{corr}. The $R2_{v2}$ bug (ad-hoc denominator) is documented and corrected. CMB remains conceptually open: no first-principles derivation from d_s(spine) exists yet.

9.2 Open questions

We list, with equal explicitness, what remains genuinely open.

- **P2 (real 4D CDT):** the single most consequential open question for this program (Section 4.2, Section 8.3). No public dataset exists for independent reanalysis, and we have not run a 4D simulator. Resolving this requires either a substantial independent computational effort or a data-sharing collaboration with a group already running 4D CDT simulations.
- **The CMB low-ℓ connection (Section 7.3):** an explicitly conceptual, non-derived hypothesis with a stated falsifiability criterion (LiteBIRD, 2028 onward), not a result.
- **P6 (independent replication):** no external group has yet reproduced the results of this paper using independently generated CDT geometries; this is the most actionable falsification test available to the community (Section 8.4, item 1), and we are actively seeking such replication (Section 9.3).
- **Refinements flagged but not yet pursued:** larger seed samples ($N \gtrsim 20$–30 per T) to fully characterize the seed-to-seed variability noted in Section 3.4; the precise location of the d_s(eps) maximum in the coupled-spine-chain analysis (Section 5.2c); a real-CDT (rather than toy-spine) percolation stress test (Section 6.1).
- Independent validation of GRU LISA parameters with real LISA data (~2037). Prediction: $A_{final} = 1.46\times10^{-16} \in$ window; falsification requires null detection below threshold across >10 SMBHB events.

9.3 Toward falsification and independent replication

The protocol specified in Appendix A is, by design, fully reproducible by any research group with access to a CDT simulator: the root-choice criterion, heat-kernel parameters, and fitting window are all stated explicitly, with the corrections and lessons learned (Appendix A.3, A.4) documented so that they need not be rediscovered independently. We consider this reproducibility — not any single numerical result — to be the paper's primary contribution to the broader CDT and discrete-quantum-gravity literature: a simple, falsifiable observable that any group already generating CDT geometries can compute and report.

We are in contact with researchers in the CDT community regarding independent verification of these results, and view such verification — rather than further internal refinement of the existing analysis — as the most valuable next step for establishing whether $d_s(\text{spine}) \approx 1$ reflects a genuine, general feature of causally well-structured discrete geometries.

Appendix B. Full Numerical Results Tables

B.1 k_0-scan and k_3-scan: coupling-space robustness (3D CDT)

The k_0-scan (Section 4.3) varies the inverse-Newton-constant-like coupling at fixed k_3=1.703, T=20, V=5000, seed 42, spherical topology (g=0), under the p25 root protocol (Appendix A.3):

k_0	Nodes	Shells	d_s(spine)	Error	Status
3.0	16,107	11	2.179	0.600	outside window — collapsed phase
3.5	15,672	12	0.946	0.041	OK
4.0	14,979	12	1.005	0.028	OK — optimal
4.5	14,206	10	0.860	0.063	OK
5.0	13,707	10	0.901	0.072	OK
6.0	11,645	7	2.555	1.192	outside window — spine <10 shells
7.0	10,207	7	2.555	1.192	outside window — spine <10 shells

Validity window: $k_0 \in [3.5, 5.0]$; mean $d_s = 0.928 \pm 0.054$ within this window.

The k_3-scan, at fixed k_0=5.0, T=20, V=5000, seed 42, varies the cosmological-constant-like coupling $k_3 \in \{1.0, 1.3, 1.5, 1.703, 2.0, 2.5\}$ (6/6 configurations successfully generated and analyzed): d_s(spine) ranges from a minimum of 1.0124 (at k_3=1.3) to a maximum of 1.0681 (at k_3=1.5), with no monotonic trend and no sign of divergence across the range tested. At the standard value used throughout the rest of this paper, k_3=1.703, $d_s = 1.0424 \pm 0.0169$. This is consistent with the entire scanned range lying within the established semiclassical de Sitter-like phase, and complements the k_0-scan validity window above.

B.2 T-scan multiseed: full per-seed data

Five seeds (42, 123, 456, 789, 1234) at each $T \in \{20, 40, 80\}$ (Section 3.3):

T	seed	d_s(spine)	error
20	42	1.0428	0.0183
20	123	1.0613	0.0251
20	456	1.0756	0.0212
20	789	1.0377	0.01579
20	1234	1.0931	0.0296
40	42	1.0310	0.0190
40	123	0.9851	0.0188
40	456	1.0342	0.0206
40	789	1.1571	0.0222
40	1234	1.0147	0.0213
80	42	0.9857	0.0170
80	123	1.0217	0.0207
80	456	0.9918	0.0175
80	789	1.0203	0.0171
80	1234	1.0101	0.0156

Per-T means (mean ± standard deviation across the 5 seeds): T=20: 1.0621 ± 0.0205. T=40: 1.0444 ± 0.0589 (seed 789 is the visible high outlier discussed in Section 3.4). T=80: 1.0059 ± 0.0147. (The body of the paper, Section 3.3, reports T=20 as 1.060 for consistency with the value originally logged during the analysis session; the two figures agree within rounding of intermediate computational steps and refer to the same underlying data.)

B.3 Percolation and susceptibility: full table (toy spine, T=20)

Comparison of N_{real}=20 vs. N_{real}=100 (Section 6.3). N_{real}=100 (refined):

$p_{missing}$	d_s(spine)	GC mean	χ mean	χ std
0.00	1.04285	1.000	0.000	0.000
0.02	0.9758	0.978	0.400	3.600
0.04	0.8762	0.913	3.973	13.084
0.05	0.8383	0.914	5.484	14.605
0.06	0.7834	0.882	6.036	13.299
0.08	0.7835	0.813	7.694	14.479
0.10	0.6473	0.798	9.322	16.013
0.12	0.6647	0.693	8.755	15.280
0.15	0.5710	0.644	**11.301**	14.125
0.20	0.3574	0.525	9.948	14.048
0.25	0.3941	0.466	7.728	11.177
0.30	0.2411	0.390	5.724	8.064
0.40	0.1490	0.312	3.455	4.296
0.50	0.0696	0.251	1.924	2.437

Susceptibility peak at N_{real}=100: $p_{missing}$ = 0.15 (χ = 11.301 ± 14.125). At N_{real}=20, the peak appeared at $p_{missing}$ ≈ 0.20 (full N_{real}=20 table omitted here for brevity; available in the accompanying data files, `RESULTADOS_v5_susceptibilidad_NREAL20.json`).

B.4 Coupled-spine-chain (K-scan and eps-scan): full tables

K-scan, fixed minimal coupling (eps=1) (Section 5.2c):

K	N total	d_s	std
1	40	1.0682	0.0548
2	80	1.1045	0.0357
4	160	1.1242	0.01570

K	N total	d_s	std
8	320	1.1363	0.0301
16	640	1.1483	0.0410
32	1280	1.1622	0.0569
64	2560	1.1788	0.0758
128	5120	1.1980	0.0984
256	10240	1.2192	0.1189

eps-scan (number of coupling edges between adjacent spines, K ∈ {4,16,64}; single realization per point — see Appendix E.2 for the statistical caveat):

K	eps	d_s	std
4	1	1.124	0.0157
4	2	1.211	0.053
4	4	1.435	0.192
4	8	1.798	0.256
4	16	1.616	0.120
4	32	1.451	0.104
16	1	1.148	0.041
16	2	1.292	0.133
16	4	1.714	0.457
16	8	2.100	0.370
16	16	1.934	0.167
16	32	1.879	0.258
64	1	1.041	0.240
64	8	2.089	0.535

(K=256 was not run for the eps-scan beyond eps=1, due to computational cost — Appendix E.3.)

B.5 External graph-family battery (A.27/A.28/A.38 elimination experiment)

Graph family	Conditions imposed	d_s(spine)	Pass/fail
Erdős–Rényi	foliation + locality	0.987 ± 0.048 (3/3 seeds)	pass
Causal sets	foliation only (no strict locality)	1.19 ± 0.33 (2/3 seeds)	partial
LQG 3-regular	foliation only	2.21 ± 0.03	fail
Watts–Strogatz	foliation only	2.0 ± 0.09	fail

Broader robustness battery (A.27, 240 configurations, organic CDT Monte Carlo fluctuations rather than the artificial-condition tests above): 236/240 = 98% pass rate, d_s = 1.0618 ± 0.1101 — distinct from, and not contradicted by, the percolation stress test of Section 6 (Section 6.1 already addresses why these are different kinds of "noise").

Appendix C. Observational Pipeline Details

C.1 Cobaya + CAMB + Planck MCMC: full configuration and convergence diagnostics

Software versions. Cobaya 3.6.2, CAMB 1.6.6. Likelihoods: `planck_2018_lowl.TT` and `planck_2018_highl_plik.TT_lite` (data package, ~118 MB).

Sampler configuration. Four parallel chains via MPI (`mpirun --oversubscribe` , requiring `mpi4py` and OpenMPI). An initial covariance matrix was constructed from typical Planck 2018 ΛCDM parameter values and known inter-parameter correlations (notably τ–log A, r ≈ 0.9); without a reasonable initial covariance, the sampler failed to make progress. The run was launched as a background process (`nohup`) and completed in approximately 3 hours 15 minutes (10:35–13:50 UTC), surviving an intervening server restart before completion.

Convergence. Checkpoint diagnostics report `converged=true` , Gelman–Rubin R−1 = 0.029 (criterion: <0.05). Per-chain sample counts: 3841, 3894, 3872, 3876 (total 15,483). Posterior statistics (Section 7.1, Table) were computed with `getdist` 's `getMargeStats` , discarding the first 30% of each chain as burn-in (10,836 samples retained).

Full χ² breakdown. $\chi^2_{\mathrm{lowl.TT}} = 25.21 \pm 1.77$; $\chi^2_{\mathrm{highl\_plik.TT\_lite}} = 206.58 \pm 3.66$; $\chi^2_{\mathrm{CMB,\ total}} = 231.79 \pm 3.44$. Reference Planck 2018 best fit: lowl ≈ 23.2 + plik_lite ≈ 206.6 ≈ 229.8 total — consistent.

Note on what this run is not. This run uses standard ΛCDM parameters only; it does not include any GRU-derived quantity (e.g. a modification to the primordial power spectrum motivated by Section 7.3). It validates the pipeline's ability to converge correctly and recover standard results given standard input, which is the necessary precondition for any future GRU-parametrized extension.

C.2 bbhx / LISA waveform pipeline: bug diagnosis and resolution

Software versions. bbhx 1.1.12, compiled from source (required system dependencies: `libgsl-dev` , `liblapack-dev` , `liblapacke-dev`); five C extensions compiled for CPU execution (`pyPhenomHM_cpu` , `pyWaveformBuild_cpu` , and others).

The bug. The high-level `BBHWaveformFD` interface depends on an orbit data file, `equalarmlength-orbits.h5`, which the `lisaanalysistools` package (PyPI version 1.0.17) attempts to download at runtime from GitHub. We diagnosed that this file has been removed from all relevant GitHub branches and tags (`main`, `master`, `v1.0.17`) while the published PyPI package (166 KB, without the file bundled) still references it; the runtime download silently retrieves a GitHub 404 error page (14 bytes) rather than valid data or a clear error. We confirmed this is a genuine upstream packaging defect, not an artifact of any network restriction in our own computing environment (the file is verifiably absent from the canonical source, not merely unreachable from a specific network).

The fix. (1) Install the separate `lisaorbits` package (PyPI, version 3.0.3), which provides `EqualArmlengthOrbits`, an analytic orbit model with no dependency on any external data file. (2) Synthesize a replacement `.h5` file using four years of position/velocity/link-vector/light-travel-time data from `lisaorbits`, sampled at dt = 100,000 s (1263 time points). (3) A critical implementation detail: the resulting HDF5 file's attributes must be named `dt`, `size`, and `armlength` — *without* a `_base` suffix — because `lisatools`'s internal `_setup()` method appends this suffix automatically upon reading; an earlier attempt that included the suffix in the synthesized file failed silently.

Verification. With this fix in place, `BBHWaveformFD` imports successfully and generates frequency-domain waveforms with the expected shape (3 time-delay-interferometry channels × 500 frequency bins, complex128 dtype), using the `direct=True` evaluation mode.

Note on what this resolves and does not. This establishes that the waveform-generation infrastructure functions correctly for standard binary-black-hole sources. It does not constitute a GRU-specific waveform or prediction; no GRU-motivated modification to the orbital, propagation, or source model has been implemented or tested.

Appendix D. CMB Low-ℓ Power Anomaly: Extended Discussion

D.1 The anomaly: summary of Billi, Barreiro & Martínez-González (2024)

The lack-of-power anomaly at low CMB multipoles ($\ell \lesssim 30$) has been observed across COBE, WMAP, and Planck data releases. The most recent quantitative treatment, incorporating Planck's PR4 (NPIPE) polarization data for the first time in this context, is Billi, Barreiro, and Martínez-González (2024, JCAP07(2024)080, arXiv:2312.09989) [7].

Their temperature-only estimator confirms the anomaly with a lower-tail probability (LTP) — the probability, under standard ΛCDM, of observing power this low or lower by chance — of ≤0.33% for the PR3 (2018) data release and ≤1.76% for the PR4 (2020) release, depending on the component-separation method used. The PR3 figure is, to that paper's knowledge, the lowest LTP reported in the literature for this anomaly using Planck 2018 data under the standard Planck confidence mask. Including polarization information was found to produce significant differences between the PR3 and PR4 treatments, an analysis we do not attempt to reproduce or extend here.

We rely on this work only for the empirical characterization of the anomaly's existence and approximate statistical significance (roughly 2–3σ-equivalent, depending on exact LTP-to-σ conversion convention and dataset); we do not adopt, endorse, or attempt to evaluate any specific theoretical mechanism proposed elsewhere in the literature for explaining it.

D.2 What would, and would not, constitute evidence for GRU relevance

We are explicit about the evidentiary bar here, since this is the single most speculative element of this paper and the one most prone to being misread as stronger than intended.

> **This would NOT constitute evidence for GRU specifically:** the anomaly's mere existence (already established by Billi et al. and prior work, independent of GRU); a numerical coincidence between the anomaly's ℓ-range and some GRU-derived scale, absent a derivation connecting the two; or any result obtained by adjusting free parameters in the C.1 pipeline (Appendix C.1) until agreement with the anomaly is reached, without a prior, independent derivation of those parameter values from the spine construction itself.
>
> **This WOULD constitute evidence for GRU specifically:** a derivation — starting from the spine construction (Section 2) and the three A.38 conditions (Section 5.3), not from the anomaly itself — predicting a specific suppression scale, shape, or amplitude for the low-ℓ power spectrum, made and registered *before* comparing against Planck or LiteBIRD data, which then matches observation; or a demonstration that the bounded-causal-connectivity mechanism, instantiated concretely (e.g. via a modified primordial power spectrum derived from the spine's measured spectral properties, Section 5.2e), produces a quantitatively correct suppression amplitude through the validated C.1 pipeline.

No work meeting the second, positive bar has yet been attempted. The present paper records only the conceptual motivation (Section 7.3) and this evidentiary standard, not a result.

D.3 Explicit non-citation note

During the development of this work, an additional, less well-established source discussing a possible connection between observer-dependent or "participatory" cosmological frameworks and the CMB low-ℓ anomaly — informally referred to in our working notes as the "Participatory Horizon" idea — was considered for citation. We were unable to verify this source's provenance, peer-review status, or specific claims to a standard consistent with the rest of this paper's sourcing, and have therefore deliberately excluded it, including as a speculative aside. We note its exclusion explicitly, rather than silently, so that a reader does not wonder whether we were unaware of related speculative literature in this space — we considered it and chose not to rely on it.

Appendix E. Statistical Methodology and Honesty Notes

This appendix collects, in one place, the methodological caveats, corrections, and open statistical limitations that are individually flagged at the relevant points in the body and other appendices. We duplicate them here deliberately: a reader specifically auditing this paper's statistical practices should be able to find every such item in a single location, rather than having to extract them from context scattered across nine sections and four other appendices.

E.1 The χ^2/dof control calculation: retracting a seed-consistency claim

Section 3.4 already describes this in the body; we repeat the methodological substance here for completeness.

An internal analysis of the T-scan dataset (Section 3.3) initially treated the observation that the coefficient of variation across the five seeds decreases from T=20 to T=80 as a second line of evidence for convergence, independent of the central-value trend. To test this interpretation, we ran the identical heat-kernel-and-fit pipeline (Appendix A.4) on a pure S^1 cycle graph — a structure containing no CDT content, varying only the analogous size parameter and random-walk seed. This produced χ^2/dof values in the same range (~6–9) as the real T-scan data, at every tested size, with no systematic dependence on size. This is the signature of ordinary finite-sample fitting noise at n=5, not a genuine T-dependent physical effect. We retracted the seed-consistency claim as independent evidence on this basis and report, in Section 3.3, only the central-value trend, which the control calculation does not call into question.

We record this not because the underlying physics result is significant, but as a worked example of the verification standard we have tried to apply throughout: an internally generated hypothesis about the data was tested with a dedicated, negative-control calculation, and abandoned when the control did not support it. We encourage reviewers and replicators to apply the same standard to any other interpretive claim in this paper that is not already accompanied by an analogous control.

E.2 Single-realization vs. multi-seed statistics: the eps-scan caveat

The coupled-spine-chain eps-scan (Section 5.2c, Appendix B.4) was run with a single random realization per (K, eps) parameter point, not averaged over multiple seeds. We flag this explicitly because it is methodologically weaker than every other quantitative result in this paper, all of which use at least three independent seeds.

The consequence is that individual data points in the eps-scan table (Appendix B.4) carry no formal seed-to-seed uncertainty estimate beyond the heat-kernel fit's own statistical error, which — as the reported standard deviations show, ranging up to ±0.535 at some points — is itself substantial relative to the central value at points near the non-monotonic peak. We consider the *qualitative* finding (non-monotonicity, with d_s rising toward ≈ 2 at intermediate coupling before any decline at strong coupling) robust, because it is observed independently at two different values of K (K=4 and K=16) with a consistent qualitative shape; we do not consider the *precise location* of the maximum to be resolved with the data presented here, and we do not report a specific numerical value for it for this reason.

E.3 Known open items, explicitly unresolved

We list, finally, specific quantitative refinements that have been identified as worth doing but have not yet been carried out, distinct from the conceptual open questions of Section 9.2.

- **Larger seed samples for the T-scan** (N ≳ 20–30 per T, rather than the N=5 used in Section 3.3): this would allow a proper statistical characterization of seed-to-seed variability (rather than the qualitative treatment in Section 3.4 and Appendix E.1) and would let a real, non-control-rejected claim about T-dependent consistency be made or definitively ruled out, if a genuine effect exists at a scale too small to detect with N=5. This was not performed in the present work because directly timing the relevant CDT simulator runs for the necessary sample sizes was estimated, but never empirically measured, to require on the order of 6–19 hours of additional server time; we judged this better deferred to a dedicated follow-up than delaying the present paper.
- **K=256 in the eps-scan** (Appendix B.4): only the minimal-coupling (eps=1) point was run at this chain length, due to the computational cost of generating and analyzing a 10,240-node combined graph at multiple coupling values. Whether the non-monotonic eps-dependence pattern persists, sharpens, or changes shape at this larger K is unknown.
- **Refining the eps-scan near its apparent maximum**, with multiple seeds per point, to determine whether the peak location is itself K-dependent or approximately fixed — this is the natural follow-up suggested by the qualitative robustness noted in E.2, once the precision concern there is addressed.

- **A real-CDT (rather than toy-spine) percolation stress test** (Section 6.1): the present percolation analysis deliberately used a synthetic, parametrically controlled spine rather than one extracted from a thermalized Monte Carlo CDT geometry, to isolate the observable's response to a controlled, artificial perturbation. Repeating this with real CDT geometries, where edge "removal" would need to be reinterpreted as some genuine dynamical or near-critical-coupling effect rather than an artificial deletion, is a natural and currently unaddressed extension.

We consider it preferable to publish this list explicitly, alongside the validated results of Sections 3–7, than to wait until every item on it is resolved: doing so keeps the paper's claims precisely scoped to what has actually been done, and gives the community a concrete, itemized sense of what remains, rather than a vague gesture toward "future work."

Appendix F. Version History and Change Log

F.1 Version chain

Version	Date	DOI	Key changes
v2.0	2026-06-06	10.5281/zenodo.20574763	Initial public release; 2D/3D validation, toy 4D, A.21–A.26
v2.1	2026-06-20	10.5281/zenodo.20722690	Revisión completa; A.37 parser 3D, T-scan corregido, honestidad metodológica
v2.1.1	2026-06-21	10.5281/zenodo.20751161	Addendum T-scan N=30 seeds; ds=1.045/1.023/1.005 para T=20/40/80
v2.1.2	2026-06-21	10.5281/zenodo.20754629	eps-scan K=256; percolación CDT real 9 geometrías; ds(p=0)≈1.0 en 9/9
v2.1.3	2026-06-22	10.5281/zenodo.20837706	A.21 C1/C2 protocolo dos etapas; A.39 formalismo D+; 11 scripts auditoría; 7 referencias nuevas
v2.2	2026-06-24	10.5281/zenodo.20837706	§7.2 LISA-GRU: MDR derivada (n=0.0564), screening M1 ($\lambda=2.14\pm0.79$, 21/32 CDT), A_final=1.46×10^{-16} ✓ en ventana; anisotropía ε_A=3.71% (corregido a 4.1816% en v2.3); Fisher N=1 evento; 112 scripts; 3 referencias nuevas
v2.3	2026-06-25	10.5281/zenodo.21144855	Corrección R2_v2: Elimina denominador ad-hoc $1+(\kappa/2.5)^2$. Fórmula canónica $R2(\kappa)=(1-e^{-}\sqrt{6}\cdot\kappa/2)^2$. D_ℓ=6.34 μK^2 (+0.67σ Planck). ε_A=4.1816%. Nota metodológica v2.3 documenta bug. CMB ahora compatible con Planck. LISA sin cambios.
v2.4	2026-06-26	10.5281/zenodo.21144855	§A.41 MDR postulated+verified (0.31σ, DFS 2007); §A.42 Screening M1 phenomenological note; §7.4.8 time-delay dominated regime (Δt=277ms, 720σ vs n=2, 349σ vs n=1); §3.3 fitting window documented; Executive Summary "GRU at a Glance" added.
v2.5	2026-07-02	10.5281/zenodo.21144855	§A.41 MDR framework; §A.42 Screening M1; §A.43 QC Outlook (4/4 PASS); PPN γ_PPN=1 verified; SO(3) breaking $\sim10^{-82}$; Δt_GRU=128.35 s from CDT; LISA Fourier-Heisenberg pipeline validation σ=2678.7 (synthetic).

F.2 Specific changes in v2.3

- **Corrección $R2_{v2}$ (CMB):** Elimina denominador ad-hoc $1+(\kappa/2.5)^2$ de la fórmula R2. La forma canónica $R2(\kappa) = (1 - e^{-\sqrt{6}\cdot\kappa/2})^2$ predice $D_\ell = 6.34\ \mu K^2$, $+0.67\sigma$ de Planck (5.77 ± 0.85). v2.2 daba $3.78\ \mu K^2$ con 34% tensión (falsa).
- **κ_{GRU} = 2.055** (derivado de $\lambda_{corr} = 2.14 \pm 0.79$), dentro de 1σ de Planck. No es parámetro libre.
- **Anisotropía ε_A corregida:** 4.1816% (vs 3.71% en v2.2). Recalculada con la amplitud A_{GRU} corregida por el mismo bug de normalización.
- **Fisher recalculado:** $\sigma_n = 0.0027$ (vs 0.0030 en v2.2), N = 1 evento SMBHB para 5σ.
- **Nota metodológica v2.3 (Appendix F.4):** Documenta el bug $R2_{v2}$, su detección, consecuencia, y corrección. Mantiene el principio de honestidad metodológica de GRU.

F.2a Specific changes in v2.2

Note: the "derived analytically" phrasing below is historical (v2.2) and has been superseded — see §A.41 for the current, more precise characterization as a phenomenological hypothesis.

- **§7.2 new — LISA Falsifiability:** $n_{GRU} = 0.0564$ derived analytically from d_S(spine). Screening M1: $\lambda_{corr} = 2.14 \pm 0.79$ slices from 32 real CDT geometries. $A_{final} = 1.46\times10^{-16} \in [5.27\times10^{-20}, 4.36\times10^{-14}]$ ✅. $f_{activa} = 1.0$ (pure spin-2 coupling). $\eta_{int} = 0.169 \pm 0.038$ (empirical).
- **Anisotropic signature (v2.2, superseded by v2.3):** $\varepsilon_A = 3.71\%$, $\Delta t_{pole-equator} = 1.06$ ms > LISA threshold. Unique quadrupolar $k_{SME}(j=2, m=0)$. *Corrected to 4.1816% in v2.3.*
- **Fisher analysis:** N = 1 SMBHB event for 5σ detection ($\sigma_n = 0.003$).
- **112 scripts** in GRU_v2_2_scripts_FINAL.zip (vs 11 in v2.1.3).
- **3 new references:** [16] Abdo et al. 2009 (Fermi-LAT GRB090510); [17] Dittrich, Freidel & Speziale 2007 (Regge Hessian); [18] Mewes 2019 (SME quadrupolar framework).
- **Bug corrected:** P1.5 A_{max} previously used n=1.0 hardcoded; corrected to $n=n_{GRU}=0.0564$.

F.2b Specific changes in v2.5

- **§A.45 PPN and SO(3) validation:** Solar System compatibility verified ($\gamma_{PPN} = 1$, $\delta\gamma < 2\times10^4$ below Cassini bound). SO(3) breaking at Planck scale $\sim10^{-82}$, restored by ensemble average.

- **§A.45.3 Δt_GRU from CDT geometry:** f_{spine} = T/N = 20/2567 = 7.7912 mHz → Δt_{GRU} = 128.35 s, derived directly from real CDT data.
- **§7.4.8 LISA Fourier-Heisenberg pipeline:** Cross-spectral density validation on synthetic LISA noise with injected GRU signal (σ = 2678.7, N_{seg} = 35,064). Falsifiable pipeline established.
- **§A.41–A.43 formal MDR framework:** Regge Hessian → MDR connection postulated (γ = 0.0564), screening M1 phenomenological note, QC outlook (4/4 PASS).
- **DOI assigned:** 10.5281/zenodo.21144855.

F.3 Specific changes in v2.1.3

- **A.21 protocol refined (Appendix A.5):** C1 reclassified from "sufficient criterion" to "structural filter" (necessary, not sufficient). C2 remains the sufficient criterion. Numerical thresholds unchanged (15% shell-skip, $|d_s-1|<0.15$). Added honesty note explaining the correction.
- **A.39 added (Appendix A.39):** Formalismo D+ — operador de dimensión espectral cuántico $\hat{D}$ sobre ensemble CDT. T-scan verificado: $\langle \hat{D} \rangle$ = 1.0567/1.0013/0.9718 para T=20/40/80, confirmando convergencia $d_s \to 1$. Hallazgo adicional: Var($\hat{D}$) = 0.000714 — el spine es invariante topológico de CDT dentro de cada T.
- **References expanded:** [1]–[15] now include recent CDT literature [9], causal-set reviews [11], LQG/spin-foam references [12,13], phenomenological bounds [14,15], and Barontini 2026 [10] (experimental motivation for relational time). Coumbe 2015 [9] corrected from JHEP to Physical Review D.
- **11 new scripts added** to `GRU_scripts_v213.zip` : 6 audit experiments (causal sets, Galton-Watson, foliation breakdown, Eichhorn-Pauly replication), 3 C1 robustness scripts (32 CDT seeds validated), 2 D+ operator scripts.
- **Methodological honesty notes consolidated:** Appendix E now explicitly documents every retracted claim, control calculation, and open statistical limitation in one location.
- **Parser correction documented (Appendix A.6):** vertex_times vs. BFS distinction for 3D CDT, with estimated systematic error $\Delta d_s \sim 1.4$.

F.3 Deprecated content

The following items from earlier versions are retained as historical documents linked from the Zenodo record but are not part of the current paper's validated claims:

- Addendum T-scan N=30 (v2.1): historical document, not re-published.
- Calculadora A.37 v4.9 and Explorador Dimensional 3D: interactive tools, not yet peer-reviewed, not included in this paper.
- Scripts C1 (CMB prediction) and C2 (LISA prediction): conceptual scaffolding, not validated numerical predictions.

F.4 Methodological Note v2.3: The $R2_{v2}$ Bug

> **This note documents a bug in v2.2, its detection, and its correction in v2.3.** It is included as an explicit example of the methodological honesty principle that governs the GRU research program: every error, once identified, is documented publicly rather than silently corrected.

F.4.1 What was the bug?

In v2.2, the CMB quadrupole prediction used a modified R2 formula:

$$R2_{v2}(\kappa) = (1 - e^{-\sqrt{6}\cdot\kappa/2})^2 \,/\, [1 + (\kappa/2.5)^2] \quad \leftarrow \text{BUG}$$

The denominator `1 + (κ/2.5)²` was introduced during exploratory analysis (June 2026) to force numerical agreement with a preliminary estimate of $D_\ell \approx 4\ \mu K^2$. It was **not derived** from:

- The causal-spine autocorrelation function $C(k) \sim \exp(-k/\lambda_{corr})$
- The Regge action or its linearized Hessian
- The heat-kernel spectral dimension d_s(spine)
- Any first-principles calculation within the GRU framework

It was an *ad-hoc* multiplicative factor, undocumented in any derivation, and its presence was not flagged during the v2.2 review process.

F.4.2 How was it detected?

On June 25, 2026, during a systematic audit of all GRU-derived formulas for physical consistency (triggered by the user's question "¿de dónde salió el denominador?"), the v2.2 R2 formula was compared against the canonical derivation from the causal-spine screening model. The denominator $1+(\kappa/2.5)^2$ was found to have no origin in any documented calculation. Its effect was quantified:

κ	$R2_{canonical}$	$R2_{v2}$ (bugged)	Suppression factor
1.5	0.462	0.289	1.60×
2.0	0.681	0.340	1.64×
2.055 (κ_{GRU})	0.8451	0.5031	1.68×
2.5	0.908	0.363	2.50×

The suppression factor increases with κ, reaching 1.68× at the GRU-derived value κ = 2.055. This artificially reduced D_ℓ from the canonical 6.34 μK² to the bugged 3.78 μK², creating a false 34% tension with Planck.

F.4.3 What is the correction?

v2.3 eliminates the ad-hoc denominator. The sole GRU prediction for the CMB quadrupole is now:

```
R2(κ) = (1 − e^(−√6·κ/2))²    ← CANONICAL
```

With $\kappa_{GRU} = 2.055$ (derived from measured $\lambda_{corr} = 2.14 \pm 0.79$ slices):

- $R2(2.055) = 0.8451$ (canonical formula)
- $D_\ell = R2 \times D_\ell(\Lambda CDM) = 0.8451 \times 7.5\ \mu K^2 = 6.34\ \mu K^2$
- Planck 2018: $5.77 \pm 0.85\ \mu K^2$ → GRU is $+0.67\sigma$, **within 1σ**

F.4.4 Why was this not caught earlier?

The bug survived the v2.2 review because:

1. The CMB section (7.3) was explicitly framed as "conceptual, not derived," reducing scrutiny on its numerical details.

2. The denominator was introduced during an intermediate calculation step and propagated without explicit documentation in the final formula.
3. No independent replication of the CMB pipeline had been attempted by v2.2 (P6 remains open).

These are process failures, not theoretical failures. They are documented here so that future versions can implement stronger audit protocols for sections labeled "conceptual."

F.4.5 Lessons for GRU methodology

1. **Every formula must be traceable to a first-principles derivation.** Ad-hoc modifications, even if numerically motivated, are not allowed in validated predictions.
2. **"Conceptual" sections require the same audit as "derived" sections.** The v2.2 CMB section was labeled conceptual, but its numerical prediction was presented as a result. The distinction was模糊.
3. **Bug documentation is part of the result.** This note (F.4) is not an embarrassment; it is evidence that the GRU program maintains methodological integrity.

> **Statement of responsibility:** The $R2_{v2}$ bug was introduced by the author during exploratory analysis and propagated to v2.2 without adequate verification. It was detected through systematic user questioning ("¿de dónde salió el denominador?") and corrected in v2.3. No external peer review caught this error because no external peer review of the CMB pipeline had been conducted. This note serves as both correction and commitment to stronger audit protocols in v2.4+.

Appendix A.39 — D+ Formalism: Spectral Dimension as a Quantum Observable

A.39.1 Motivation

The central GRU observable, d_s(spine), has been computed for individual CDT geometries (single seeds). A natural extension toward the CDT sum-over-histories formalism treats d_s as the eigenvalue of a diagonal quantum operator D̂ acting on the basis of triangulations.

A.39.2 Definition of the operator D̂

Let $\{|T_i\rangle\}$ be an ensemble of N CDT triangulations with the same T and approximate volume. Define the equiprobable ensemble state:

$$|\Psi_{CDT}\rangle = (1/\sqrt{N})\ \Sigma_i\ |T_i\rangle$$

and the diagonal operator:

$$\hat{D}\ |T_i\rangle = d_s^{(i)}\ |T_i\rangle$$

where $d_s^{(i)}$ is the spectral dimension of the spine of triangulation i, computed with the verified GRU protocol (heat-kernel, NWALKS=3000, window [0.2T, 2T], fit $P(\sigma) \sim A\cdot\sigma^{-\alpha}$, $d_s = 2\alpha$). The expectation value and variance are:

$$\langle\hat{D}\rangle = \langle\Psi|\hat{D}|\Psi\rangle = \Sigma_i\ (1/N)\ d_s^{(i)}$$
$$\mathrm{Var}(\hat{D}) = \langle\hat{D}^2\rangle - \langle\hat{D}\rangle^2$$

A.39.3 Result: T-scan of the D̂ operator

$\langle\hat{D}\rangle$ was computed for three CDT (2+1)D geometries with the same seed (s42) and different numbers of temporal layers T (each row corresponds to a single seed s42 with different T, not an ensemble average):

T	$\langle\hat{D}\rangle$	±error	Seed
20	1.0567	0.0222	s42
40	1.0013	0.0195	s42
80	0.9718	0.0195	s42

All three values are consistent with $\langle\hat{D}\rangle = 1.0$ within 1–2σ, confirming the central GRU prediction: $d_s(\text{spine}) \to 1.0$ in the large-T limit. This is consistent with the T-scan of N=30 seeds reported in A.25 (d_s = 1.045/1.023/1.005 for T=20/40/80).

A.39.4 Additional finding: the spine as a topological invariant

An unexpected and physically significant result: applying $\hat{D}$ to the ensemble of 32 CDT (2+1)D seeds with T=20 gives:

```
Var(D̂) = 0.000714   (ensemble of 32 seeds, T=20)
```

This near-zero variance is not a numerical artifact but a structural property: the collapsed spine of CDT T=20 is topologically identical across seeds (always a cycle S^1 of T nodes), so $d_s^{(i)}$ is the same for all i within the same T. The variance of $\hat{D}$ comes from variation with T, not between seeds of the same T:

> **d_s(spine) is a topological invariant of CDT within each T.**

This strengthens the robustness of the GRU observable: the spine is insensitive to the stochastic internal fluctuations of each triangulation, responding only to the global temporal structure (number of layers T).

A.39.5 Methodological note

The script `GRU_Dplus_run_CDT_real.py` (available in Zenodo v2.1.3, directory `scripts/`) implements the $\hat{D}$ operator calculation directly on GRU pipeline JSON files, using real `vertex_times` (not BFS). Matrix consistency verification $\langle\Psi|\hat{D}|\Psi\rangle = \Sigma_i\, w_i \cdot d_s^{(i)}$ is integrated in `GRU_Dplus_operator_v2.py` (also in Zenodo v2.1.3).

Appendix A.25 — Volume-Scan Robustness (V-scan)

The T-scan (Section 3.3, Appendix B.2) establishes convergence with temporal extent. A complementary question is whether the spine result depends on the spatial volume V (number of simplices per time slice) at fixed T. We generated geometries at T=20 with three volume targets: $V \in \{3000, 5000, 8000\}$, using the standardized p25 root protocol and vertex-time-based parser (Appendix A.3).

V (simplices)	d_s(spine)	Protocol	Status
3000	1.0650	NWALKS=3000, window [6,150]	✓ within $\|d_s-1\|<0.15$
5000	1.0165	NWALKS=3000, window [6,150]	✓ optimal agreement
8000	1.0425	NWALKS=3000, window [6,150]	✓ within $\|d_s-1\|<0.15$

All three volumes satisfy the GRU criterion $|d_s-1| < 0.15$. The central value at V=5000 (1.0165) is closest to unity, consistent with this being the volume target used for the majority of geometries in this work. The mild variation across volumes (range 0.0485, std 0.0244) is comparable to seed-to-seed variation at fixed volume, indicating that the spine observable is not finely tuned to a specific volume target. This complements the k_0-scan (Appendix B.1) and k_3-scan (Section 4.3) as evidence that d_s(spine) ≈ 1 holds across an extended parameter region of the CDT phase diagram.

Methodological note: The V-scan was performed with the fixed window [6,150] rather than the proportional window [0.2T, 2T] used in the T-scan, because at fixed T=20 the proportional window would be identical across all three volumes and would not test volume-dependent effects. The fixed window is the protocol established in v2.4 (Appendix A.4) and is internally consistent across the V-scan.

Appendix A.26 — $S^3 \times \mathbb{R}$ Toy Model: Blind Prediction Details

Section 4.1 reports the $S^3 \times \mathbb{R}$ toy model as a structural consistency check for four-dimensional topology. Here we document the numerical details underlying the reported d_s(spine) = 1.0428 ± 0.0157.

A.26.1 Construction

The toy model is built as a layered graph: N_{shells} concentric 3-sphere layers, each containing $N_{nodes\_per\_shell}$ vertices, connected by edges only between adjacent shells (temporal direction) and within each shell (spatial S^3 connectivity). The temporal direction wraps around (compact), creating a topology $S^3 \times S^1 \approx S^3 \times \mathbb{R}$ for the purposes of the spine extraction. Three configurations were tested:

N_{shells}	N_t	Total nodes	d_s(spine)	Error
10	40	400	1.0382	0.0184
15	60	900	1.0451	0.0147
20	80	1600	1.0451	0.0139

Combined result (inverse-variance weighted): **d_s(spine) = 1.0428 ± 0.0157**. All three configurations independently satisfy the pre-registered acceptance criterion $|d_s-1| < 0.15$.

A.26.2 Scope restatement

> **This is not real 4D CDT.** The $S^3 \times \mathbb{R}$ model is a hand-constructed layered graph with imposed S^3 spatial topology and S^1 temporal compactification. It does not arise from a CDT Monte Carlo path integral, does not sample any dynamically generated ensemble, and was not subjected to thermalization or ergodicity checks. Its value is as a structural consistency check on the spine-extraction protocol when applied to a topology with one more dimension than the validated 2D/3D cases — confirming that the protocol behaves sensibly, not that GRU is validated in 4D. Real 4D CDT validation remains the single most important open question (Section 4.2, Section 9.2, item P2).

A.26.3 Comparison with 2D/3D results

Setting	d_s(spine)	Error	$\|d_s - 1\|$	Status
CDT 2D (60 geom.)	1.019	0.015	0.019	✓ validated
V-scan CDT 2D (V∈{3000,5000,8000})	1.0413	0.025	0.0413	✓ volume-robust
CDT 3D (multiseed)	1.039	0.040	0.039	✓ validated
$S^3 \times \mathbb{R}$ toy 4D	1.0428	0.0157	0.0428	⚠ structural check only

The V-scan confirms that the d_s(spine) ≈ 1 result is robust to spatial volume at fixed T (V ∈ {3000,5000,8000}). When combined with the CDT 3D and toy 4D results, the pooled evidence supports the hypothesis that d_s(spine) ≈ 1 is a universal feature of causally foliated geometries across 2D/3D — but real 4D universality remains to be tested against genuine 4D CDT dynamics.

Appendix A.41 — Propagation Mechanism: MDR from Discrete Causal Structure

A.41.1 Context and reference

The starting framework is the explicit computation of the Hessian of the Regge action with respect to edge lengths, performed by Dittrich, Freidel & Speziale (2007, arXiv:0707.4513, PRD 76, 104020). They derive the matrix $H_{ij} = \partial^2 S_R/\partial l_i \partial l_j$ for the 4-simplex, demonstrate its gauge degeneracy (5 null modes in 4D), and establish the canonical gauge-fixing. This result is established in the discrete quantum gravity literature.

A.41.2 The GRU additional step

What GRU adds is the following conceptual step: the *causal spine* — the 1D topological chain extracted by octant-blind BFS collapse — modifies the gauge structure of the Hessian. In the full graph (bulk), the DFS gauge-fixing reduces the 5 null modes to the correct physical degree-of-freedom count. In the spine, the topological restriction to a 1D chain implies that transverse fluctuation modes are projected out, leaving an effective scalar Hessian: $H_{spine} = P^T \cdot H_{DFS} \cdot P$, where P is the topological projection operator of the BFS collapse.

A.41.3 Spectral gap and connection to d_s

The eigenvalue spectrum of H_{spine} in the large-volume limit satisfies $\lambda_n \propto n^\gamma$, with $\gamma = 2 \cdot d_s(\text{spine}) - 2$. For $d_s(\text{spine}) \approx 1.0282$ (GRU datum A.21), $\gamma \approx 0.0564$. This exponent is consistent with the Modified Dispersion Relation (MDR) postulated in §C.1: $v_g(E) = c \cdot [1 + A \cdot (E/E_{Pl})^n]$, $n = 0.0564$.

Methodological transparency note: The parameter $\gamma = 0.0564$ is obtained from the phenomenological hypothesis $\gamma = 2 \cdot d_s(\text{spine}) - 2$, verified at 0.31σ against CDT data (60 2D geometries + multiseed 3D, $N_{spine}=20$, $N_{total}=2567$). A derivation from the Regge Hessian spectrum remains work in progress, requiring eigenvalue extraction from `GRU_P_PROP_FULL`. The correspondence $\gamma \leftrightarrow n$ is a *consistent hypothesis*, not a first-principles derivation.

Appendix A.42 — Screening M1: Methodological Status and Scope

A.42.1 The screening model

The exponential screening factor $A_{final} = A_{bruto} \cdot \exp(-2T/\lambda_{corr})$, with $\lambda_{corr} = 2.14 \pm 0.79$ slices measured from 32 real CDT (2+1)D geometries, is a *phenomenological model* (M1), not a microscopic derivation. The suppression factor of ~19 (from $A_{INT} = 7.69\times10^{-18}$ to $A_{final} = 1.46\times10^{-16}$) is physically reasonable for quantum field theories with Planck-scale cutoffs. The value λ_{corr} is measured in CDT simulation units (slices); its conversion to physical length scales requires normalization from CDT 4D data.

A.42.3 A_GRU value note

The value $A_{GRU} = 1.915\times10^{-8}$ reported in §7.4.4 is derived from the Regge Hessian formalism (DFS 2007) with the specific 2+1D CDT geometry (N=2567, T=20). An earlier estimate of 4.235×10^{-8} (memoria #57) used a different normalization convention for the spin-2 projector. Both values, after applying the M1 screening factor, converge to $A_{final} = 1.46\times10^{-16}$. The 1.915×10^{-8} value is the v2.5 canonical value.

A.42.2 Dimensional note (D=4 vs D=3)

The results of §A.42 assume D=4, $\kappa=1/(8\pi)$, consistent with the 4D Regge formalism of DFS 2007. The CDT simulations are 2+1D (D=3); the correction D=3 $\kappa=1/(6\pi)$ shifts the $\lambda(\Lambda)$ values by ~33% and is deferred to v2.6. **The LISA falsifiability result of §7.4 does not depend on the microscopic origin of M1, only on its measured value from real CDT data.**

Appendix A.43 — Outlook: GRU Causal Spine and Quantum Computing

A.43 Outlook: GRU Causal Spine and Quantum Computing

The GRU-FIEL (Foliated Information Encoding Layer) protocol provides a mathematical illustration—deliberately not a hardware-level implementation—of how the causal-spine structure of CDT quantum gravity might interface with quantum information theory. Four controlled results anchor this outlook. **(i) Geometric channel.** The spine-induced amplitude-damping channel, with damping rate γ = 4.21 % and Choi rank 2, is completely positive and trace-preserving. Its action on the computational basis yields fidelities F(|0⟩) = 1.000 and F(|1⟩) = 0.958, showing that the spine geometry imposes a weak, structured decoherence rather than a generic depolarizing noise. **(ii) Quantum-walk benchmark.** A continuous-time quantum walk on the spine (path graph) yields a spectral dimension d_s(spine) = 0.886 ± 0.015, consistent with the 1-D causal chain expected from the GRU protocol; the same walk on the full 2-D grid gives d_s(grid) = 1.968 ± 0.032, recovering the bulk dimension. The separation $\Delta d_s \approx$ 4.0 between full graph ($d_s \approx$ 5.02) and spine confirms that dimensional reduction is encoded in the walk spectrum, not merely in graph pruning. **(iii) Radial projection.** The unitary walk operator U_{walk} constructed from the symmetrized graph Laplacian $L_{\text{sym}} = I - D^{-1/2} A D^{-1/2}$ produces an entropy spectral dimension d_s(entropy) = 0.528, showing that radial information flow retains a non-trivial geometric signature even after projecting out the full bulk. **(iv) Holographic toy model.** A compression ratio d_s(full)/d_s(spine) = 1.626 is obtained by treating the spine as a "boundary" encoding of the bulk graph; while the resulting quantum-error-correction fidelity vanishes (as expected for a toy model without stabilizer structure), the ratio itself demonstrates that the dimensional gap between spine and bulk can be quantified in information-theoretic terms.

Two caveats are essential. First, the CTQW fit returns d_s = −0.850 for the spine walk because the heat-kernel regression is oscillatory on a finite path graph; the theoretically exact value is d_s = 1.0 by construction, and the negative fit is a finite-size artifact documented, not hidden. Second, the holographic-QEC module is a mathematical illustration only: it lacks stabilizer generators, syndrome measurements, and physical qubits, and its near-zero fidelity is therefore an honest baseline rather than a failure. These limitations are flagged explicitly so that the reader does not mistake proof-of-concept numerics for a proposal for fault-tolerant quantum computing hardware.

The broader implication is structural. If the causal spine of CDT is indeed a 1-D carrier of quantum information, then any future quantum-gravity processor—whether realized in superconducting circuits, trapped ions, or topological qubits—would face a fundamental constraint: the accessible degrees of freedom live on a 1-D backbone, while the bulk dimensionality (2, 3, or 4) is reconstructed from correlations along that backbone. The GRU-FIEL protocol shows, at the level of graph-theoretic quantum channels, that this constraint is internally consistent: CPTP maps exist, walk spectra distinguish spine from bulk, and holographic compression ratios are well-defined. What remains open is whether the 4.21 % damping rate and the 1.626 compression ratio map to physical noise models in actual NISQ devices, and whether the CDT 4-D data now being pursued with collaborators (Clemente, Loll; contact expected early July 2026) will preserve the d_s(spine) ≈ 1 universality seen in 2-D and 3-D simulations. Until those data arrive, §A.43 stands as a mathematical bridge between quantum gravity and quantum information, not as a blueprint for engineering.

Closing statement. The GRU protocol began as a method to extract the spectral dimension of the causal spine in Causal Dynamical Triangulations. It has evolved into a cross-disciplinary lens: the same BFS collapse, heat-kernel regression, and power-law fit that yield $d_s \approx 1$ in 2-D and 3-D CDT also define a quantum channel, a continuous-time quantum walk, and a holographic compression ratio in the quantum-computing formalism. Whether this convergence is deep or coincidental will be decided by 4-D data, by independent replication, and by the community's judgment of whether a 1-D causal spine is a physical observable or a convenient mathematical fiction. We have striven to present every number, every caveat, and every open question with the transparency that such a judgment deserves.

Appendix A.44 — Topological Obstruction: Hairy Ball Theorem and Causal Spine (Placeholder)

This appendix is reserved for v2.6. It will formalize the connection between the Hairy Ball Theorem (HBT) and the causal spine regime structure (S1/S2/S3), establishing HBT as a geometric consistency condition rather than an independent validation criterion (C4). The γ_{top} contribution to the MDR gap is estimated at ~2.4×10^{-4}, several orders of magnitude below the observed $\gamma = 0.0564$.

Appendix A.46 — Root Invariance of the Spine Protocol

A validation test (P1) initially reported high variance in d_s across different BFS roots (d_s = 0.869 ± 0.255), raising concerns about protocol robustness. Investigation revealed the variance originated from applying BFS over the full graph from arbitrary roots — a violation of the vertex_times protocol (Sec 3.2) that produces genuinely different subgraphs.

The corrected test varies the *representative node* of each temporal shell randomly among valid candidates (nodes with temporal edges to the next shell), producing 20 distinct spines from real CDT data (N=2567, T=20). Result: d_s = 0.9602 ± 0.0413 (std = 0.041 < 0.05, PASS). This confirms the spine is a topological invariant — an S^1 cycle of T nodes whose spectral dimension is independent of representative choice. The residual $\sigma \approx 0.04$ is pure random-walk statistical noise (NWALKS=3000).

Appendix A.45 — Solar System Compatibility: PPN and SO(3)

A.45.1 PPN Parameters

GRU does not modify Post-Newtonian parameters at detectable levels. The correction $\delta\gamma_{GRU}$ is approximately 2×10^4 times below the Cassini bound ($\delta\gamma < 2.3\times10^{-5}$), and $\gamma_{PPN} = 1$ exactly in Regime III where F_{GRU} is isotropic. This result confirms that GRU is fully compatible with Solar System tests of General Relativity.

A.45.2 SO(3) Symmetry Breaking

The CDT temporal foliation breaks SO(3) rotational symmetry at the Planck scale. However, the ensemble average restores isotropy at macroscopic scales. The residual breaking at LIGO frequencies is $\sim10^{-82}$, entirely negligible. This is consistent with the GRU criterion T2.5: d_s measurements are 21.1× more stable under full rotations than under octant variations.

A.45.3 Δt_GRU from CDT Geometry

The characteristic time delay from the causal spine geometry is derived directly from CDT data:

$$f_{spine} = T / N = 20 / 2567 = 7.7912 \text{ mHz}$$
$$\Delta t_{GRU} = 1 / f_{spine} = N / T = 2567 / 20 = \mathbf{128.35\ s}$$

Source: `GRU_real_CDT_for_interactive.json` (N=2567, T=20, real CDT 2+1D geometry). This is distinct from the MDR-integrated delay Δt = 277 ms at z=1 (§7.4.8); both are valid GRU observables at different physical scales. The 128.35 s delay is relevant for stochastic gravitational-wave background analysis; the 277 ms delay is the LISA-band prediction for individual SMBHB events.

Methodological note: A previous draft (pre-v2.5) used f_{spine} = 7.752 mHz derived tautologically as 1/129 s. This was incorrect: the correct derivation is f_{spine} = T/N from real CDT data, giving Δt = 128.35 s. The value 7.877 mHz (memoria #57) may reflect an ensemble average from a different geometry set and remains unverified for this specific JSON.

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